Electronică Analogică



Curs 3 – Joncțiunea PN. Dioda.

Tipuri de diode. Aplicații

V 2022

Disclaimer

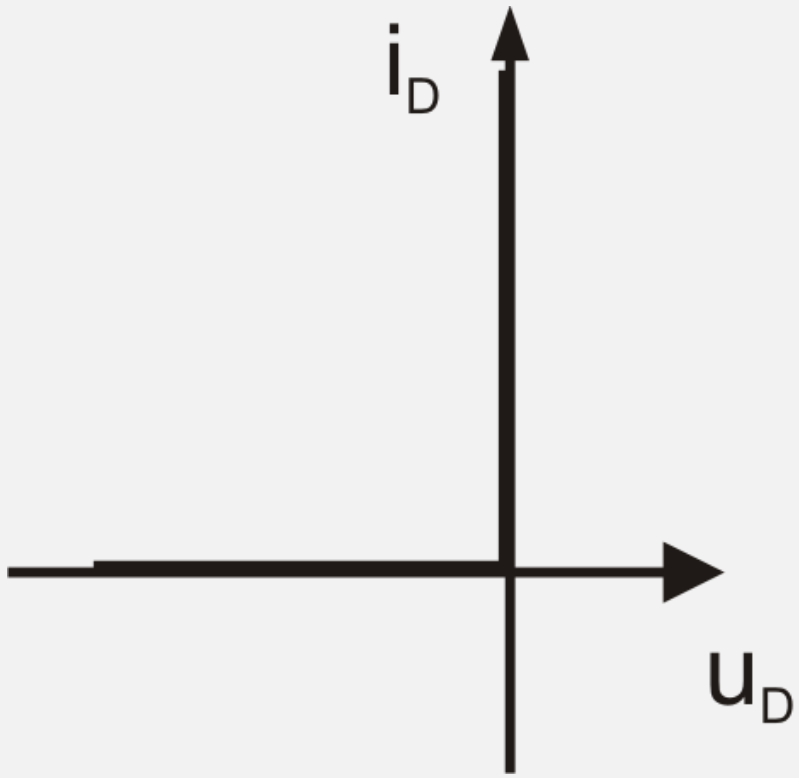
- *Slide-urile acestea reprezintă un sport de curs / rezumat.*
- *Slide-urile nu acoperă toate explicațiile și demonstrațiile oferite în timpul cursului.*
- *Slide-urile nu limitează în vreun fel materia cerută la examen.*
- *Pentru varianta completă, sunteți așteptați la curs*

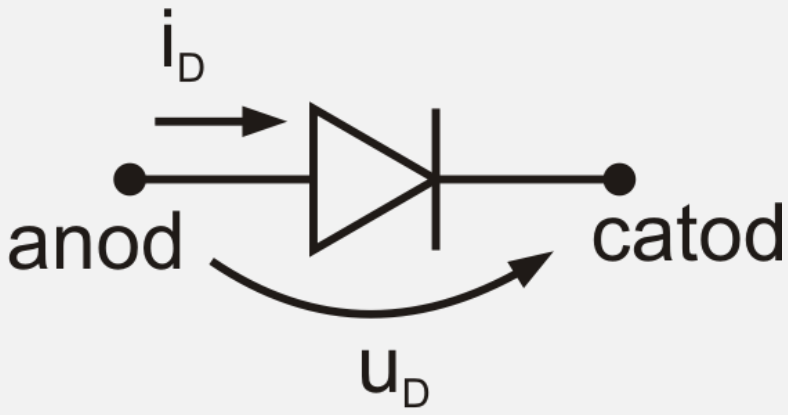
Joi 10:00 / 12:00 AN030

Luni 12:00/14:00 EC004/AN030

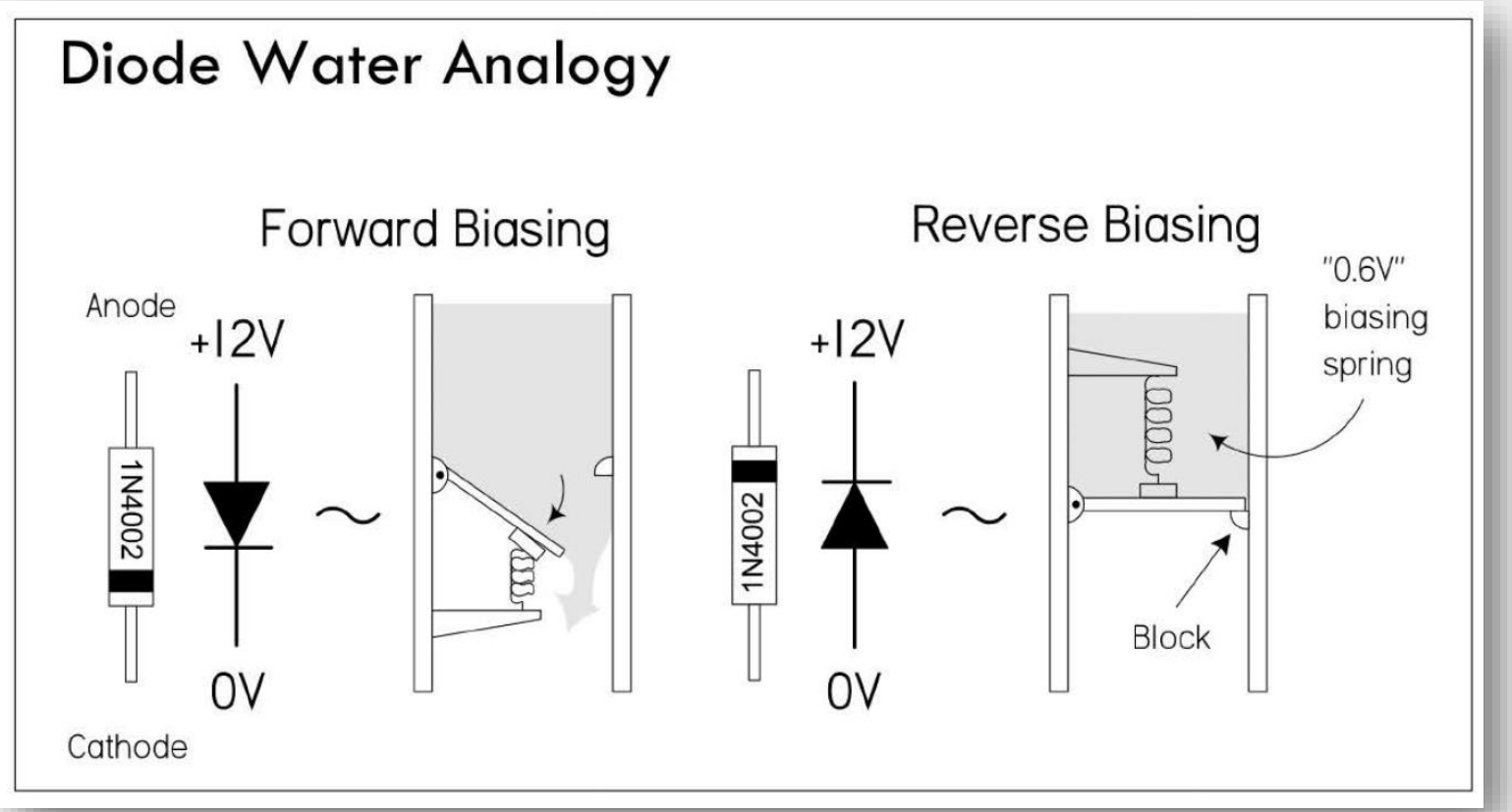
Dioda – intro (part 1 & 2)

Dioda Ideala





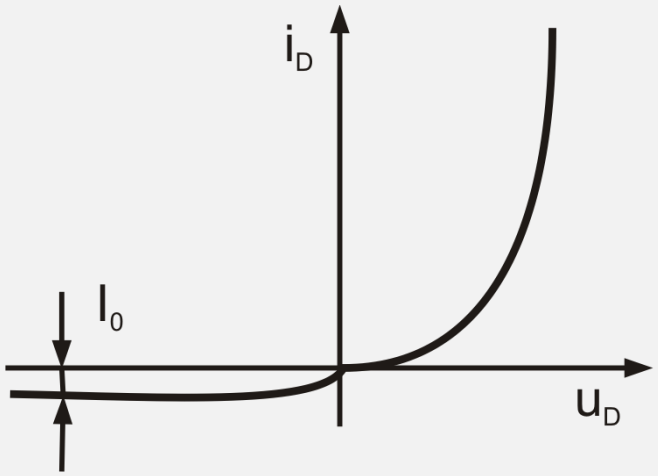
Analogie Dioda/Apă

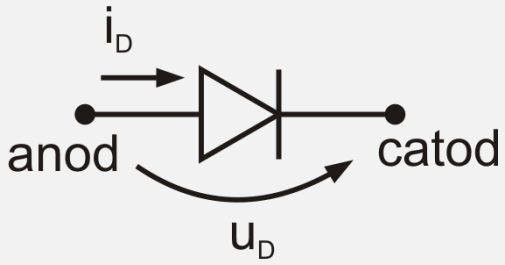


Atenție! 12V

©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

Caracteristica I-U a joncțiunii PN ideale





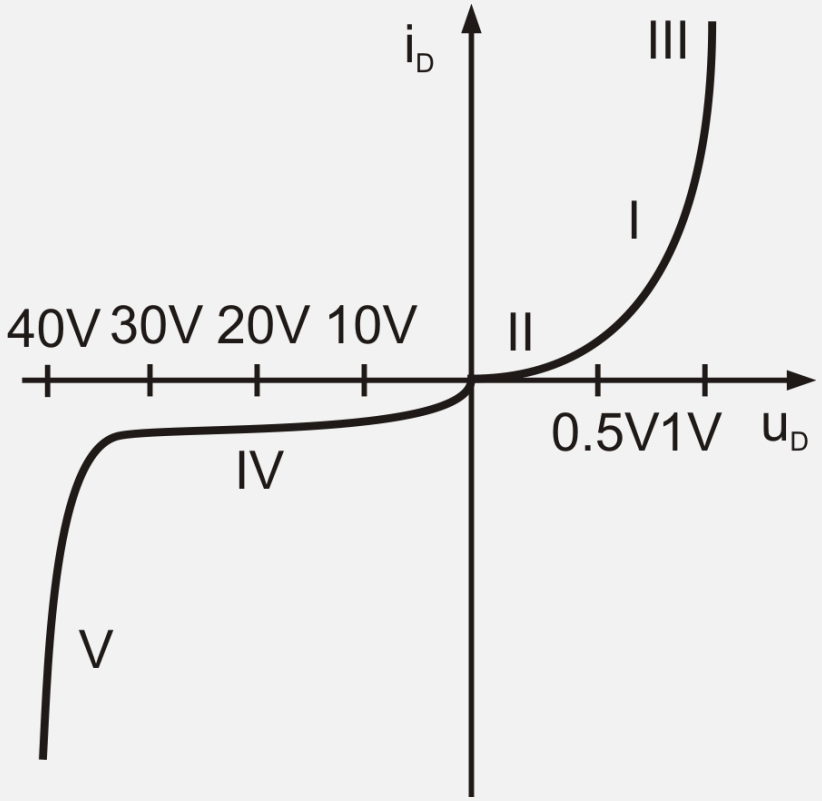
depinde de temperatură

$$i_D = I_0(e^{\gamma\frac{qu_D}{kT}} - 1)$$

caracteristică exponențială

Io- curentul de saturaţie al diodei;
u_D tensiunea de polarizare;
k - constanta lui Boltzman;
q - sarcina electronului;
T - temperatura în Kelvini;
γ - constantă.

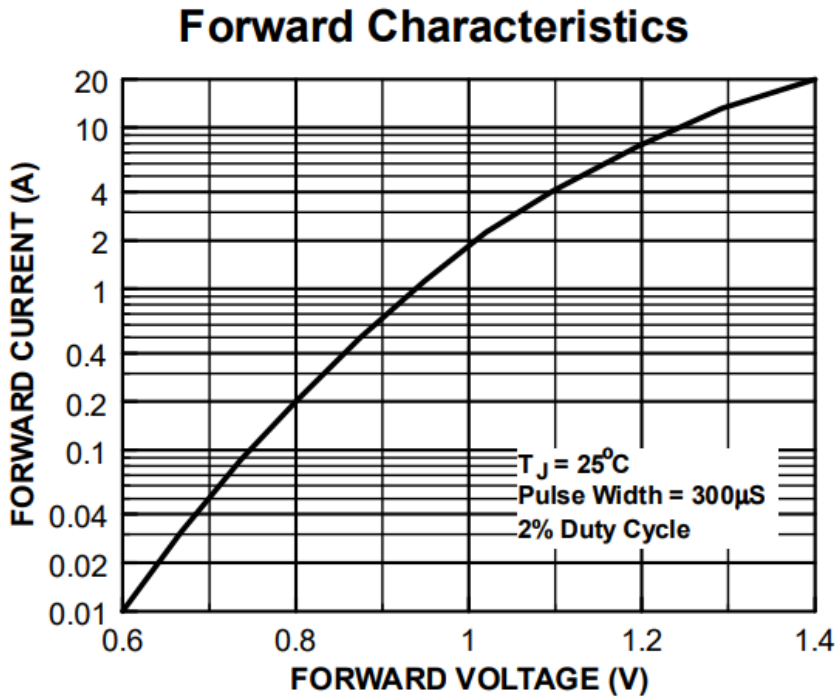
Dioda reală





DO-41

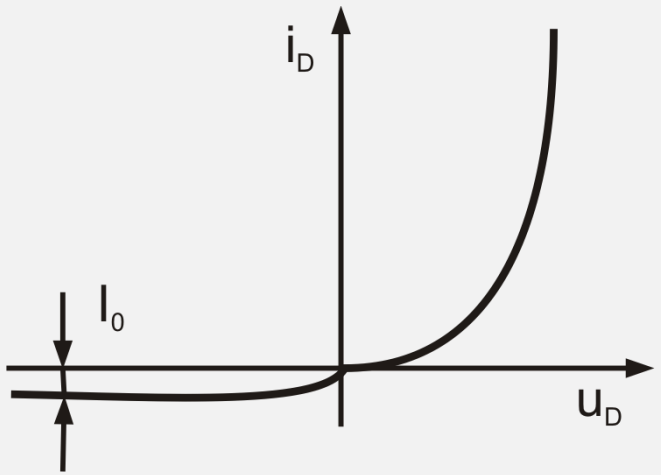
COLOR BAND DENOTES CATHODE



Caracteristica I-U a joncțiunii PN ideale

$$I_0 = qn_i^2\left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A}\right)A$$

depinde de aria joncțiunii
(în general, putere suportată mai mare
⇒arie mai mare
⇒I_0 mai mare)



A - suprafaţa joncţiunii;

ni - concentraţia intrinsecă de purtători;

Dp, Dn - constantele de difuzie;

Lp, Ln - lungimile de difuzie ale purtătorilor de sarcină;

NA, NA, - concentraţiile de impurităţi.

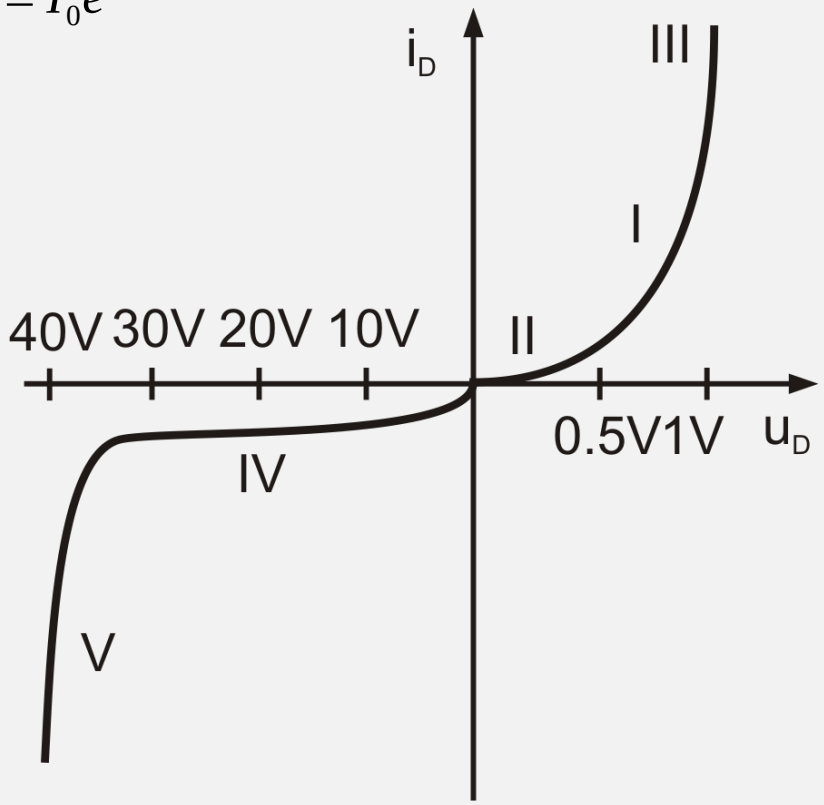
10 ÷ 100 nA pentru dioda din siliciu de mică putere
1 ÷ 100 µA pentru dioda din germaniu

Caracteristica I-U a diodei reale

- Zona I – caracteristică ideală
- Zona II – curenți de valoare mică
- Zona III – curenți mari
  - contează rez => liniarizare
- Zona IV – tensiuni inverse mici
- Zona V – tensiuni inverese mari
  - Zener
  - multiplicare în avalanșă

$$i_D = I_0 e^{\frac{qu_D}{kT}}$$

$$i_D = I_0 e^{\frac{qu_D}{\gamma kT}}$$



Noțiuni teoretice de bază



Recapitulare Chimie

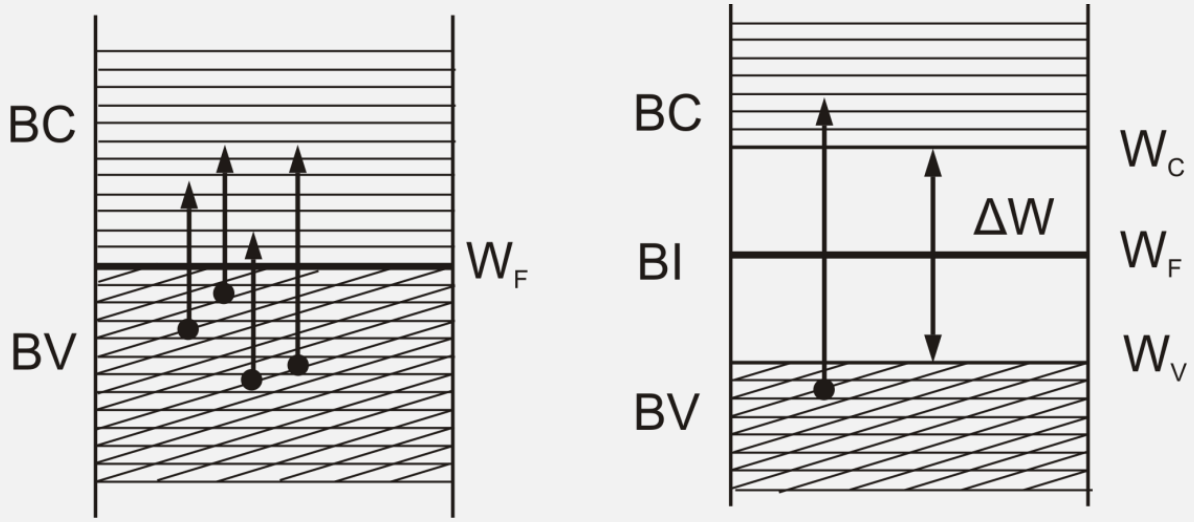
Conductori, semiconductori, izolatori

- Materialele sunt clasificate după abilitatea lor de a conduce electricitatea
- După conductibilitatea electrică corpurile solide sunt:
  - supraconductoare – R = 0
  - *conductoare* $\sigma > 10^3\ mho/cm$
    - conductibilitatea scade cu temperatura
    - exemple: argint, cupru
  - *izolatoare*
    - nici la temperaturi foarte mari nu prezintă o conductibilitate electrică important
    - exemple: sticla, teflonul (obs: div perfect G = 0)

Conductori, semiconductori, izolatori

- După conductibilitatea electrică corpurile solide sunt:
  - *semiconductoare* $\sigma \in (10^{-7}, 10^{3})\ mho/cm$
    - *conductibilitatea* depinde pronunţat de temperatură
    - exemple: siliciu, germaniu

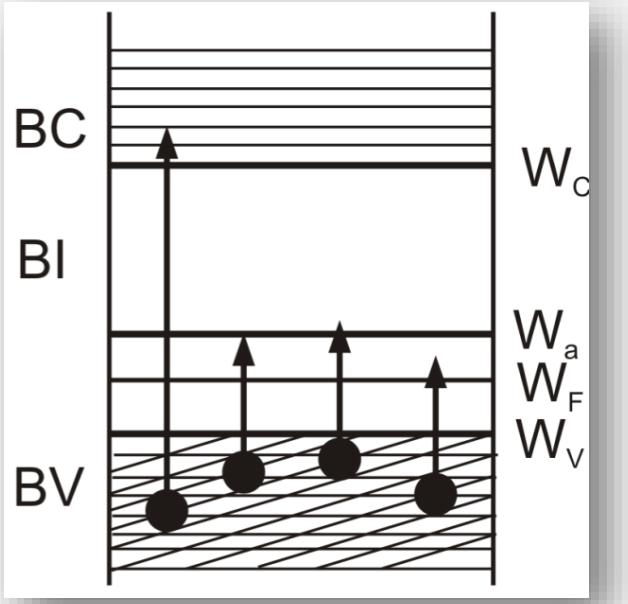
Analiză pe bază de niveluri energetice

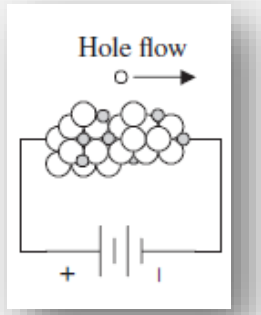


- germaniu: $\Delta W = 0{,}67$ eV
- siliciu: $\Delta W = 1{,}1$ eV
- diamant: $\Delta W = 6\text{-}7$ eV

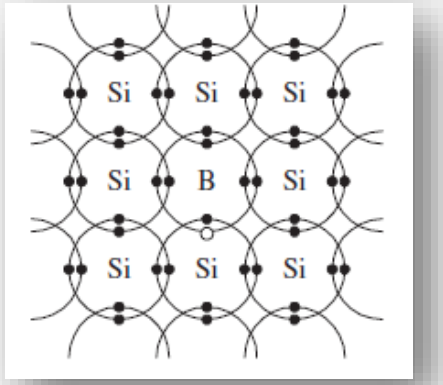
Impurificare cu elemente trivalente

- **B, Al, In, Ga**
- acceptoare
- apare uşor un gol în BV – pot participa la conducţie
- purtătorii majoritari – golurile
  - **semiconductor de tip P**





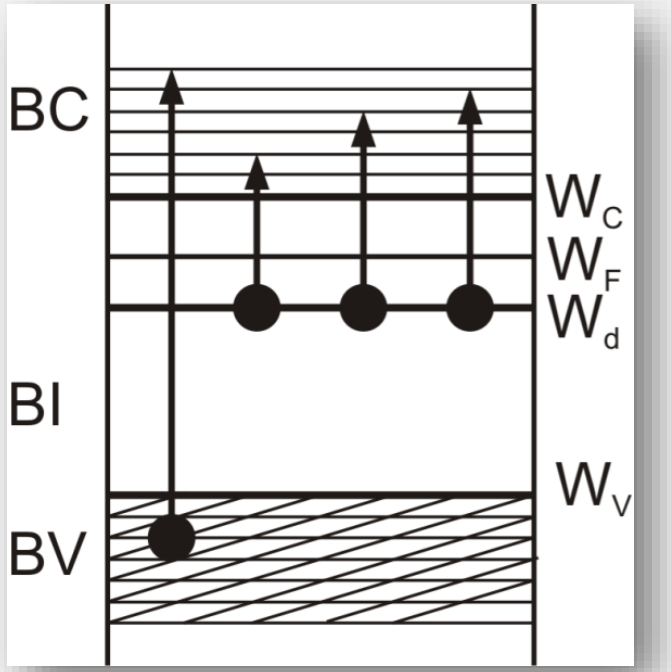
Exemplu bor (B)

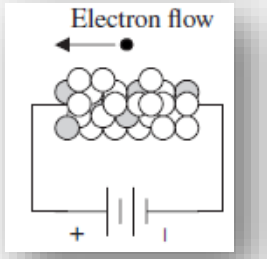


©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

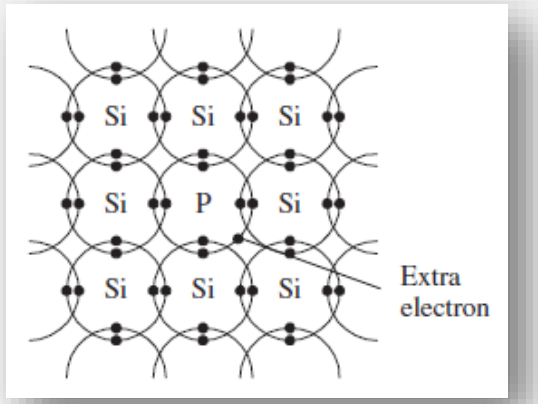
Impurificare cu elemente pentavalente

- **P, Bi, Sb, As**
- donoare
- al 5-lea electron trece uşor în BC
- purtătorii majoritari – electronii
  - **semiconductor de tip N**





Exemplu fosfor (P)



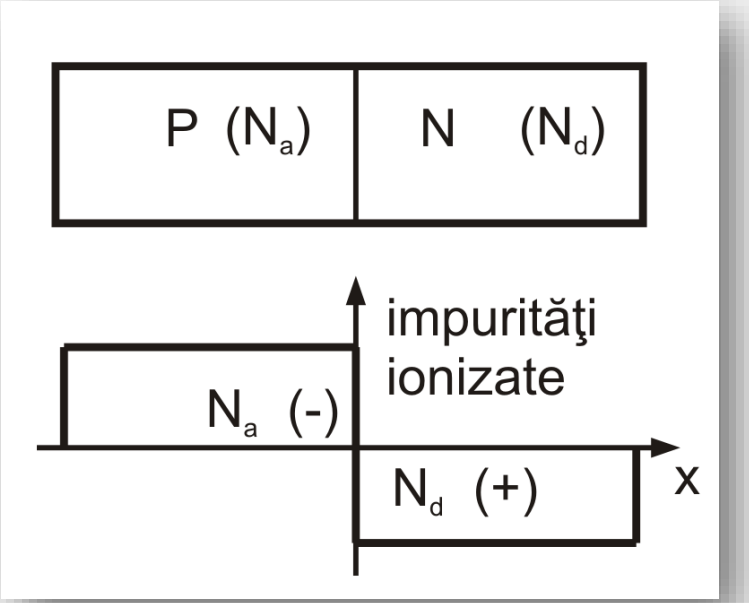
©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

Junctionea PN

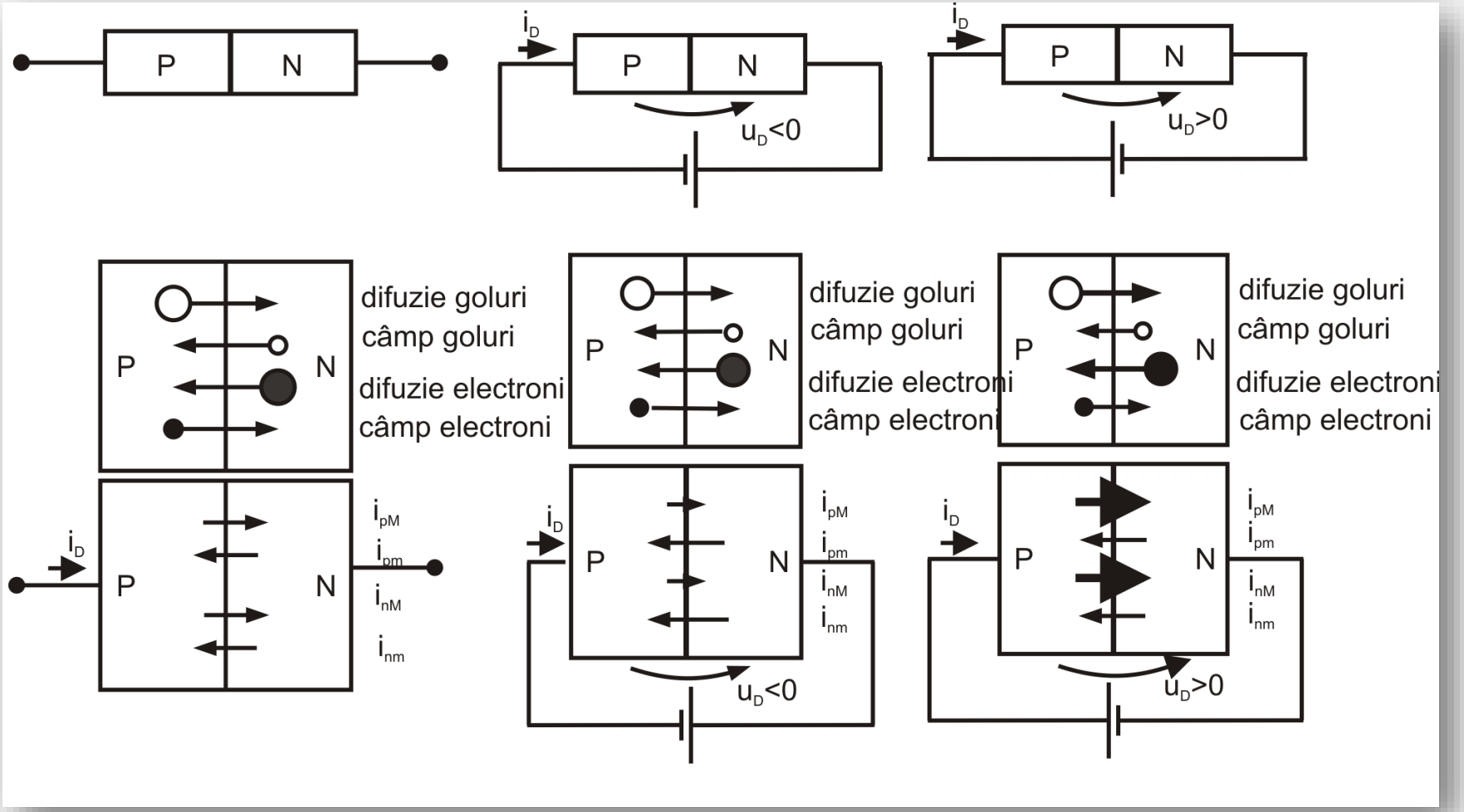
Procese fizice în joncțiunea PN ideală

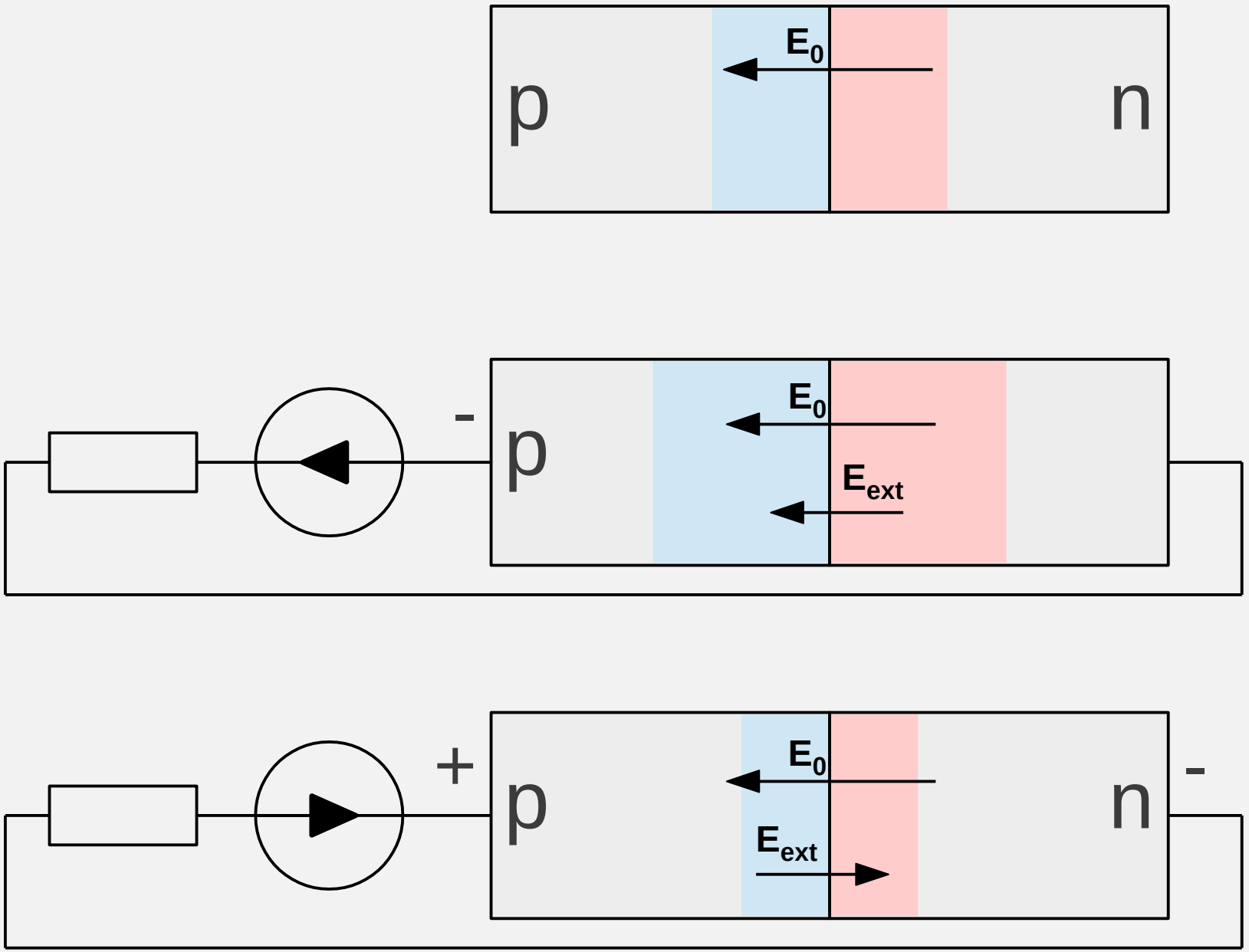
- semiconductor eterogen format din două regiuni distincte
  - P - dopată cu impurităţi acceptoare
  - N – dopată cu impirităţi donoare

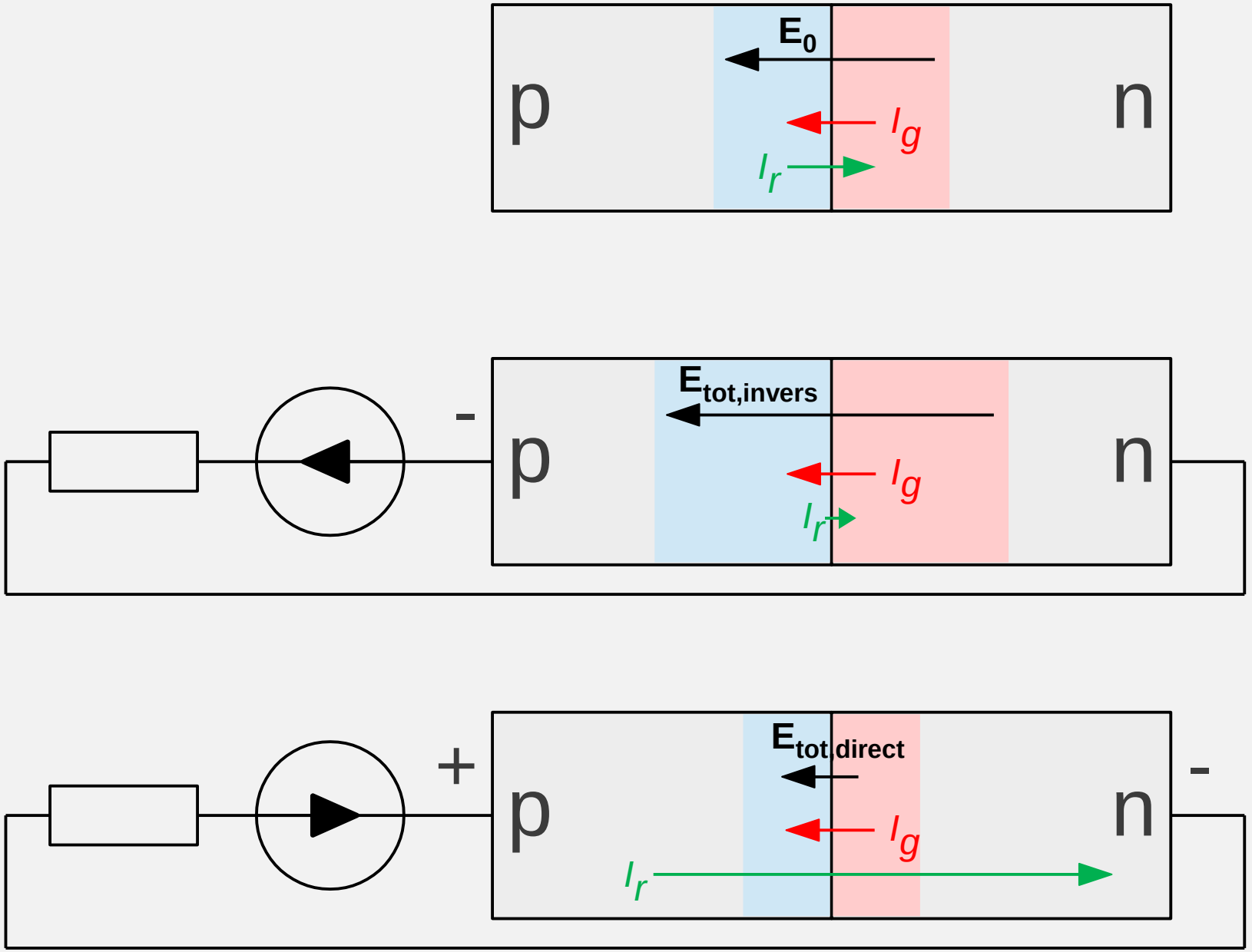
    (cele două regiuni formează o singură reţea cristalină)



Joncţiunea pn cu tensiune aplicată

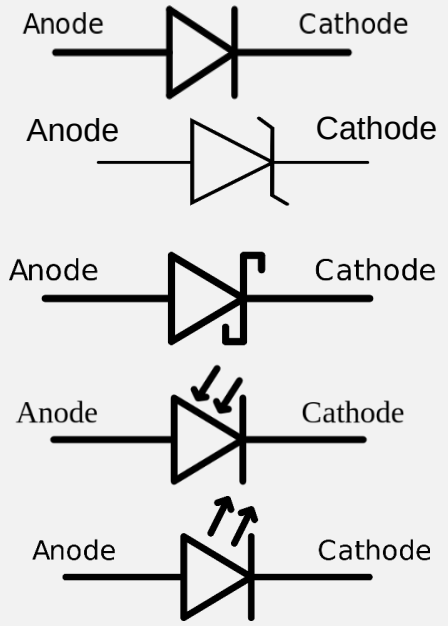



p
n
E_0
-
p
E_0
E_{ext}
+
p
n
-
E_0
E_{ext}

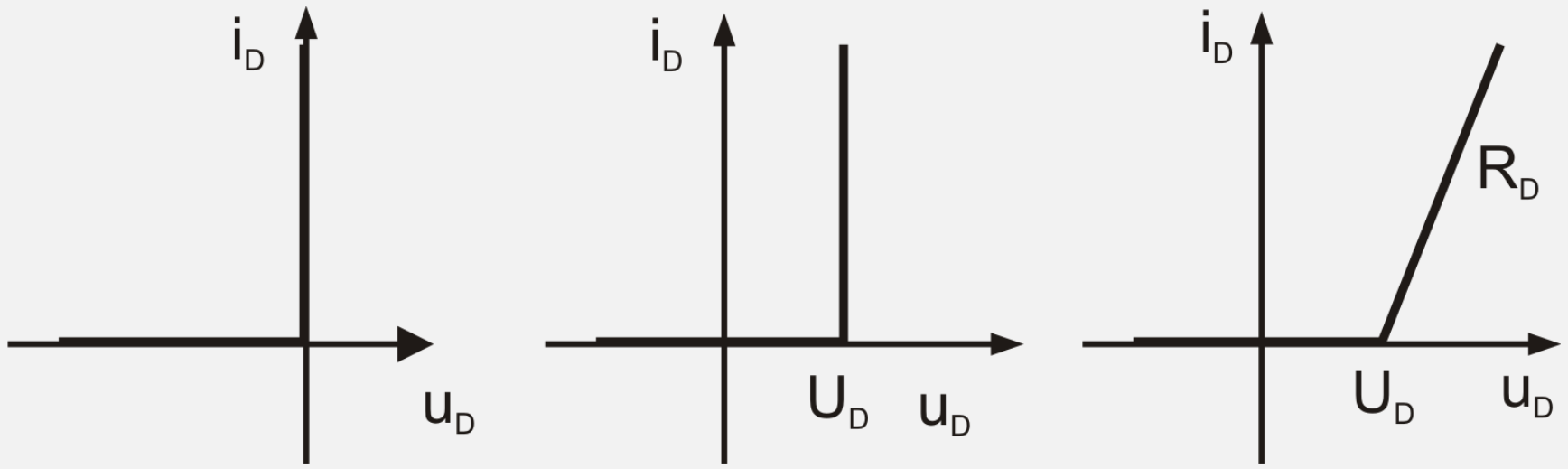

E0
p
n
Ig
Ir
-
Etot,invers
p
n
Ig
Ir
+
Etot,direct
p
n
-
Ig
Ir

Dioda

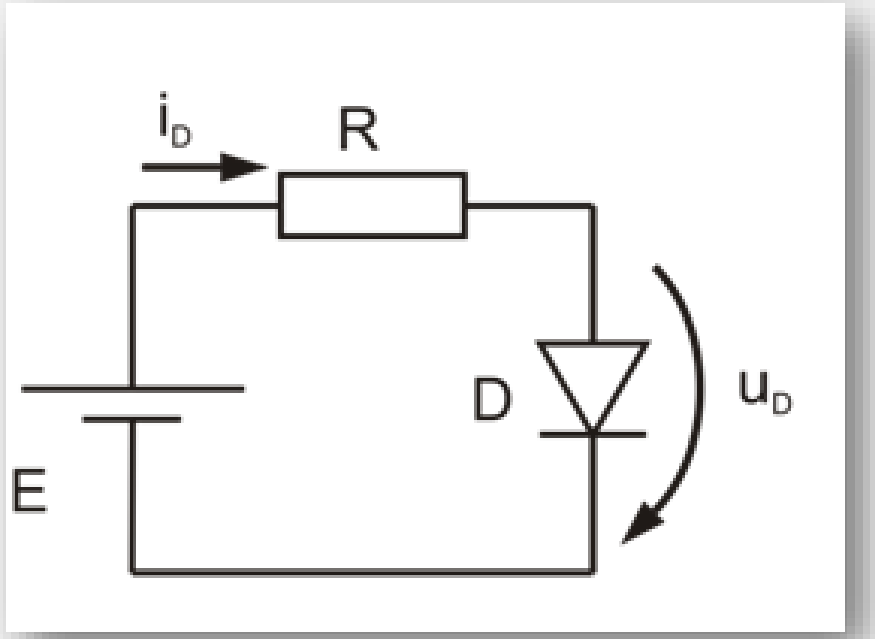
- Dioda - „one-way gate”
- Polarizare anod → catod - „forward biasing” (dioda permite trecerea curentului electric)
- Polarizare catod → anod - „reversed biasing” (dioda blochează trecerea curentului electric)
- Cele mai întâlnite tipuri de diode:
  - Dioda redresoare
  - Dioda Zener
  - Dioda Schottky
  - Foto-dioda
  - LED-ul



Modele simplificate

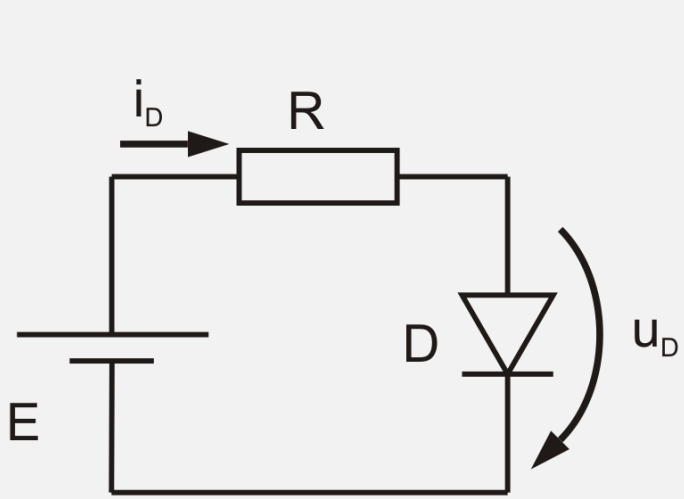


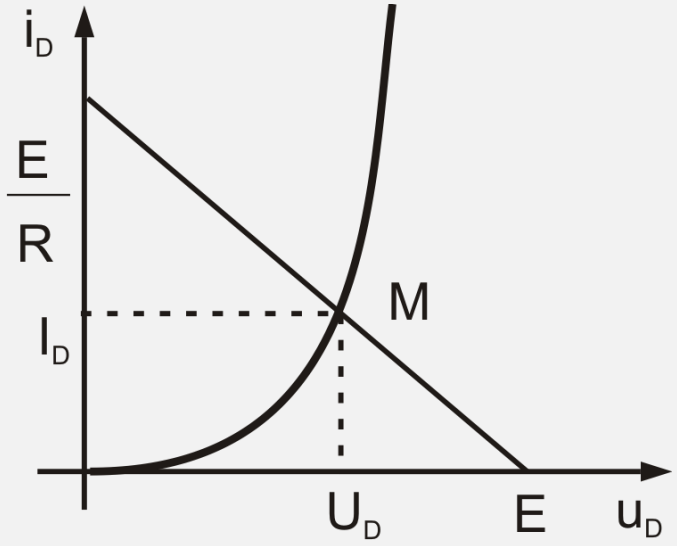
- Exemplu:



Regimul dinamic al joncțiunii PN

Circuit de polarizare

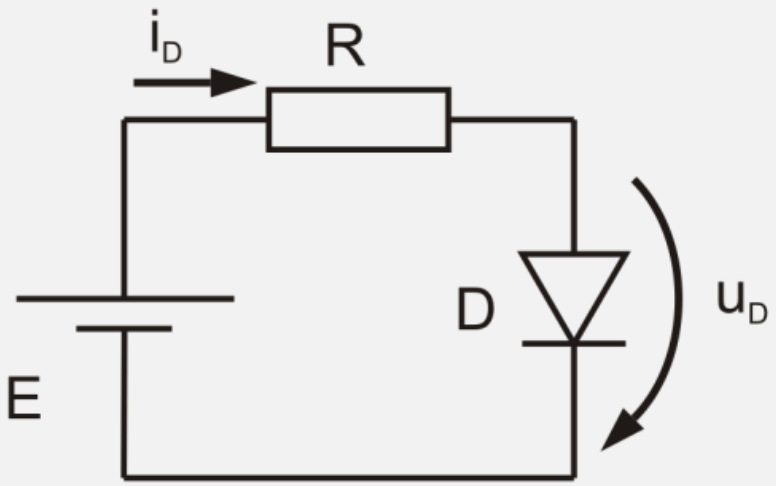


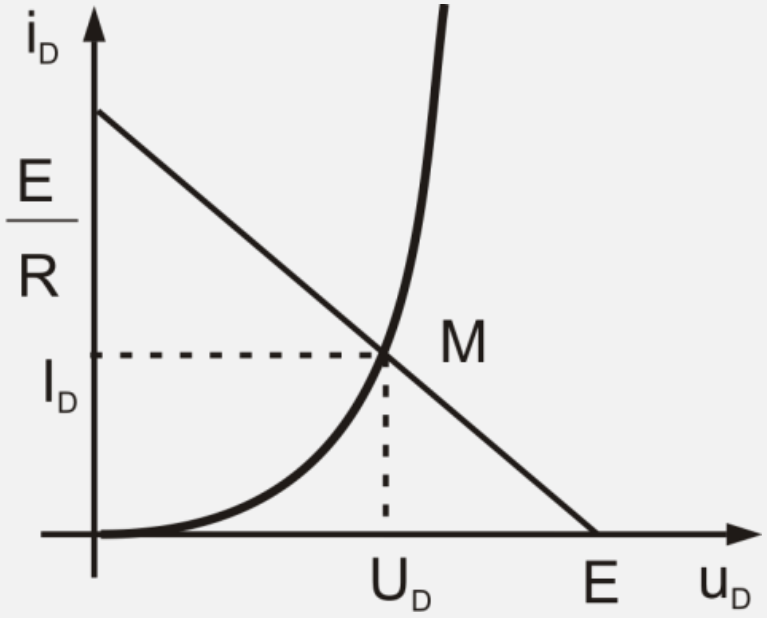


Determinarea PSF

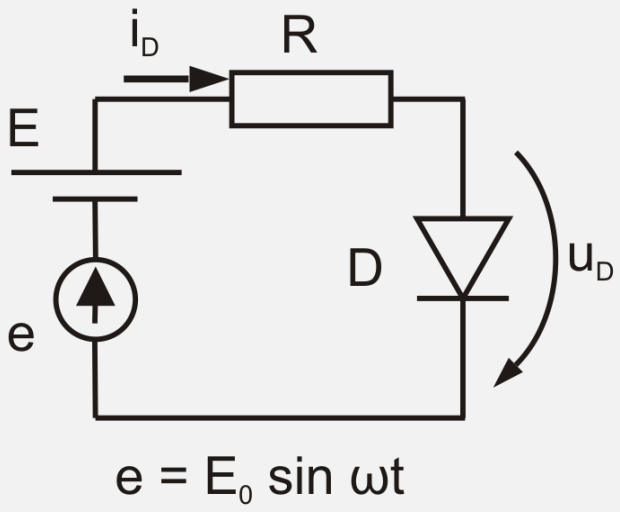
$$E = Ri_D + u_D$$

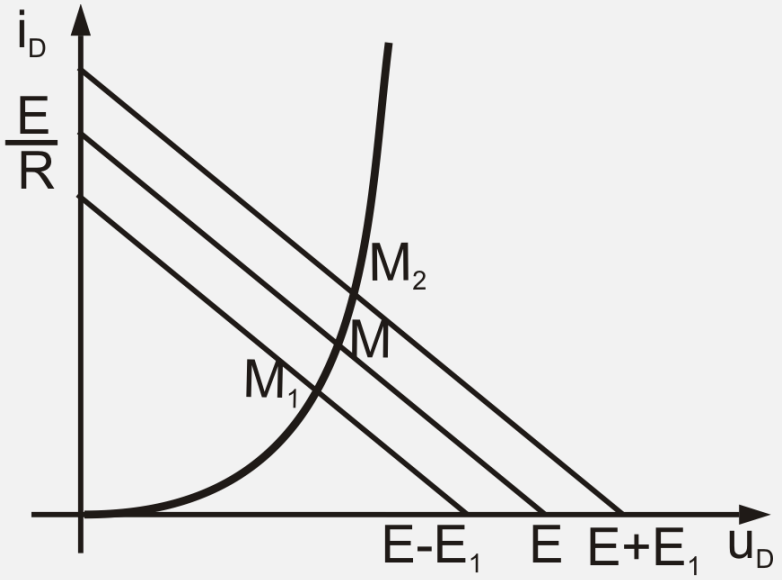
$$i_D = i_D(u_D)$$





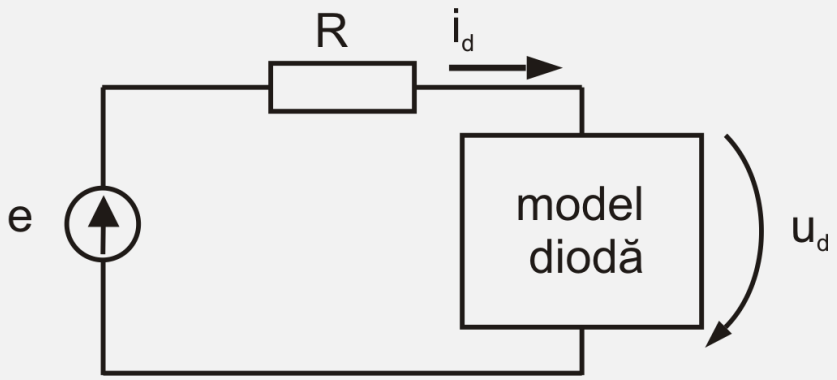
Regimul dinamic





Model diodă (1)

$$\begin{cases} u_D = U_D + u_d \\ i_D = I_D + i_d \end{cases}$$



Model diodă (2)

$$i_D = I_0\left(e^{\frac{qu_D}{kT}} - 1\right) = I_0\left(e^{\frac{q(U_D+u_d)}{kT}} - 1\right) = I_0\left(e^{\frac{qU_D}{kT}} e^{\frac{qu_d}{kT}} - 1\right) =$$

$$= I_0\left(e^{\frac{qU_D}{kT}}\left(1+\frac{qu_d}{kT}\right) - 1\right) = I_0\left(e^{\frac{qU_D}{kT}} - 1\right) + I_0\frac{qu_d}{kT}e^{\frac{qU_D}{kT}}$$

$$i_d = I_0\frac{qu_d}{kT}e^{\frac{qU_D}{kT}}$$

$$r_d = \frac{u_d}{i_d} = \frac{kT}{q}\frac{1}{I_0 e^{\frac{qU_D}{kT}}} = \frac{kT}{q(I_D + I_0)}$$

Valori tipice pentru r_d

- r_d – 1/m grafic în jurul PSF
- Exemplu:
  - pentru diodă de putere mică din Si: 26Ω la un curent direct de 1mA
- La polarizare inversă, valoare foarte mare

Capacitatea de barieră

- Determinată de sarcinile fixe rezultate în regiunea de trecere prin difuzia purtătorilor majoritari în zonele adiacente
- Efect negativ – întârzie răspunsul la frecvențe înalte

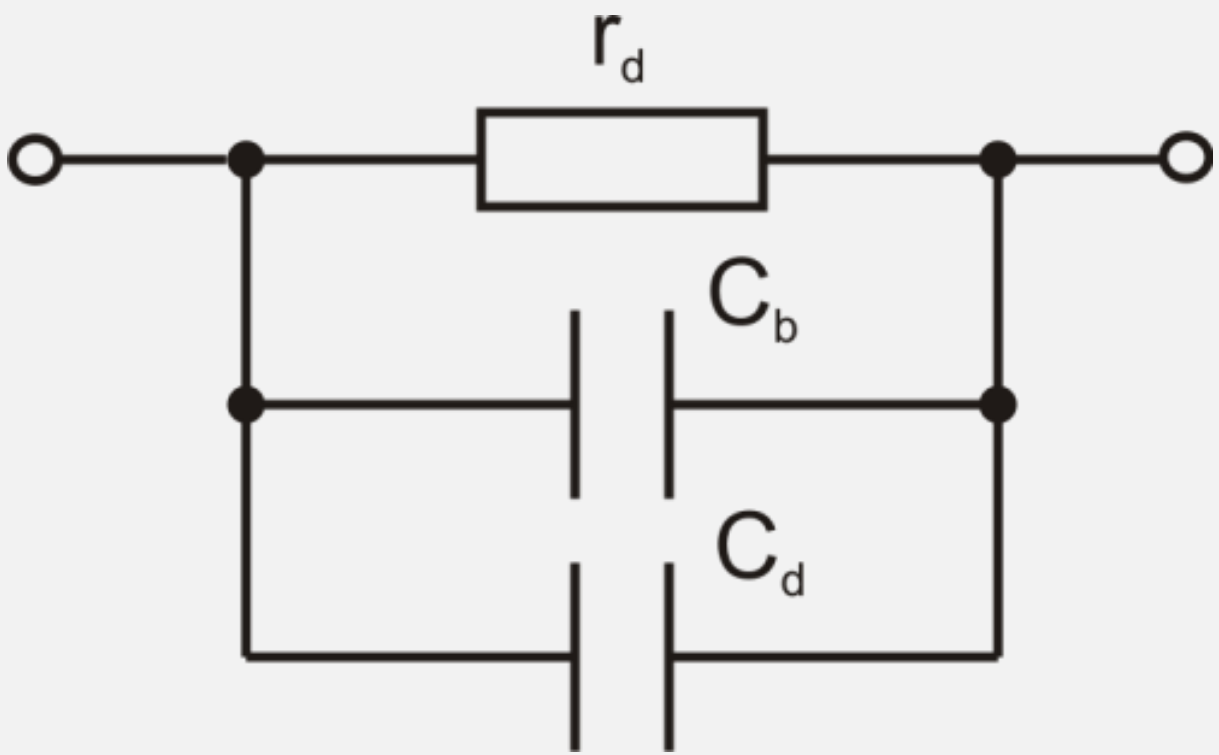
$$Q_b = -qN_a AL_p \qquad C_b = \left.\frac{dQ_b}{du_D}\right|_M$$

Capacitatea de difuzie

- Determinat de surplusul de sarcină obținut prin difuzie
- Proporțională cu curentul direct prin diodă
- Mai importantă decât capacitatea de barieră în conducţie directă
- Neglijabilă la polarizarea inversă a diodei

$$Q_d^p = qA\int_{l_n}^{\infty} p_{exces}dx \qquad C_d^p = \left.\frac{dQ_d^p}{du_D}\right|_M \qquad C_d = \frac{q}{kT}\tau_p I_D$$

Circuit echivalent pentru semnale mici și frecvențe mari



Dioda Si



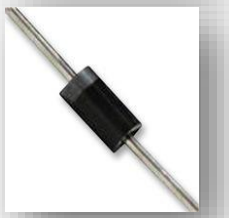
Parametri importanți

- Joncțiunea PN
- Substrat de Si

Denumire (ro)	Denumire (eng)	Valori tipice
Tensiune de prag	Knee voltage	0,6V – 1V
Curent direct maxim	Maximum Forward Current	10mA –100A
Tensiune inversă max	Max reverse voltage	1000V
Curent de staturație	Reverse current	10÷100 nA
Timp de comutare	Reverse recovery time	10ns÷10u
Putere disipată	Total power dissipation	100mW÷100W

Exemple









- 300mA / 1V (la 10mA) – 1N4148 - http://www.vishay.com/docs/81857/1n4148.pdf
- 1A / 1,1V - 1N4007 - http://www.farnell.com/datasheets/46758.pdf
- 150A / 1,17V - DSI75-12B - http://ixdev.ixys.com/DataSheet/L122.pdf
- 400A / 0.65V (la 200A)- http://www.genesicsemi.com/images/products_si/schottky/mbr40020ct_thru_mbr40040ctr.pdf

Dioda Ge



Tensiunea electrică, noduri

Caracteristici generale Ge

- Forward voltage drop – 0,3V
- I_0 - uA
- Avantaje
  - Tensiune directă mai mică
  - Potrivite pentru circuite low-power și pentru circuite radio
- Dezavantaje
  - fiabilitate mai mică
  - preț de producție mai mare
  - sensibilitate mai mare la temperaturi înalte (precauții la lipire)
  - I_0 mai mare decât pentru Si
  - nu se mai fabrică ☺

Aplicații și modele Ge

- Aplicații Radio
  - Detector AM/FM
  - Detector radio
- Aplicații TV
- Sonde RF
- Modele … EF108

Dioda Schottky



Tensiunea electrică, noduri

Parametri importanți

- Joncțiunea metal - semiconductor
- Substrat de Si tip P (modele noi – Silicon Carbide – 1000V)

Denumire (ro)	Denumire (eng)	Valori tipice
Tensiune de prag	Knee voltage	0,3V
Curent direct maxim	Maximum Forward Current	10mA –100A
Tensiune inversă max	Max reverse voltage	30-60V
Curent de staturație	Reverse current	100nA÷1mA
Timp de comutare	Reverse recovery time	100ps÷100ns
Putere disipată	Total power dissipation	100mW÷100W

Exemple







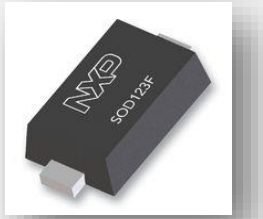
- 1A – 0,55V (la 100mA) - **MBR150 -**
  http://www.onsemi.com/pub_link/Collateral/MBR150-D.PDF
- 70mA – 0,41V (la 1mA) - **BAS70JFILM**
  http://www.farnell.com/datasheets/671292.pdf
- 2 x 27A – 1,5V (la 20A) - **C4D40120D**
  http://www.farnell.com/datasheets/671292.pdf

Exemple



- 2 x 30A; 0,47V (la 30A) -**VBT6045CBP-E3/4W** (max 45V)

http://ro.farnell.com/vishay-general-semiconductor/vbt6045cbp-e3-4w/diode-schottky-60a-45v-to-263ab/dp/2078095?in_merch=New%20Products



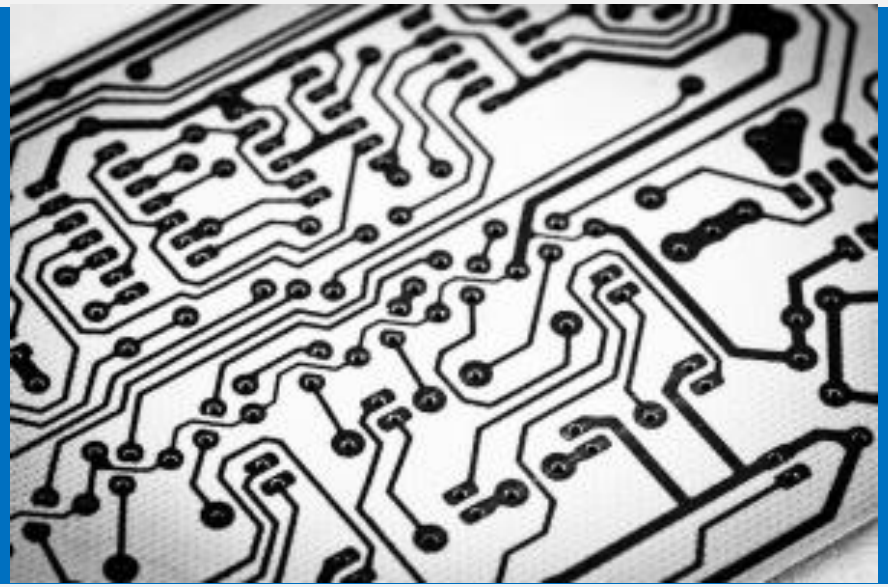
- 2 x 0,5A; 0,39V (la 0,5A) -P**MEG2005EH** (max 20V)

http://ro.farnell.com/vishay-general-semiconductor/vbt6045cbp-e3-4w/diode-schottky-60a-45v-to-263ab/dp/2078095?in_merch=New%20Products

Caracteristici generale Schottky

- Joncțiune metal – semiconductor
- Forward voltage drop – 0,15 – 0,45V
- I_0 – mai mare decât la Si
- f – până la 50GHz
- Avantaje
  - cădere de tensiune redusă
  - viteză mare de comutare (reverse recovery time scurt)
- Dezavantaje
  - tensiunea inversă redusă pentru varianta Si – metal
  - curent de saturație mare

Exemple Aplicații Diode

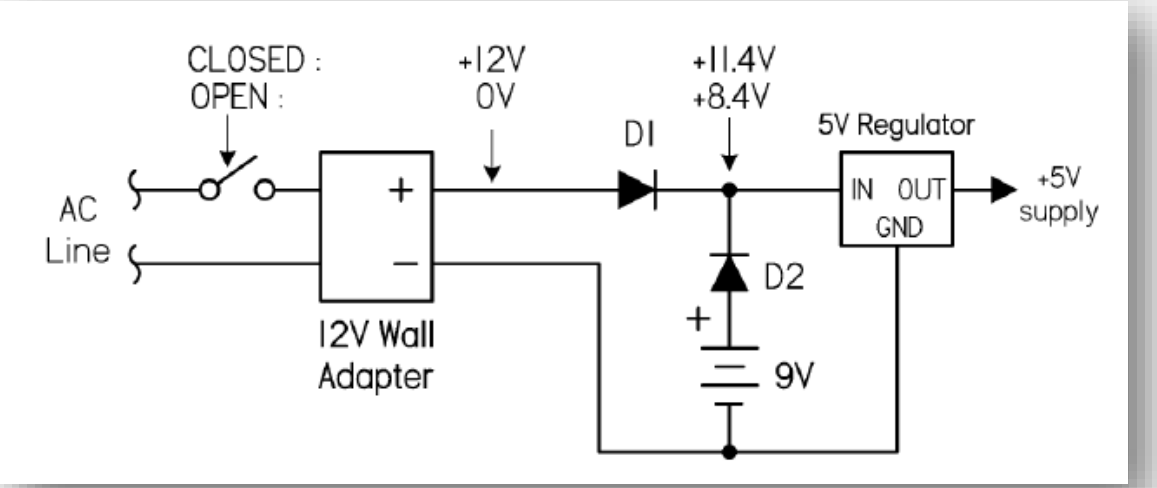


Aplicații diode

- Redresor
- Voltage clamping
- Prevenirea intrării în saturație a unui TBIP (tranzistor bipolar)
- Protecție la curent invers
- Redresare pentru surse în comutație
- AC/DC power supply

Aplicații (1)

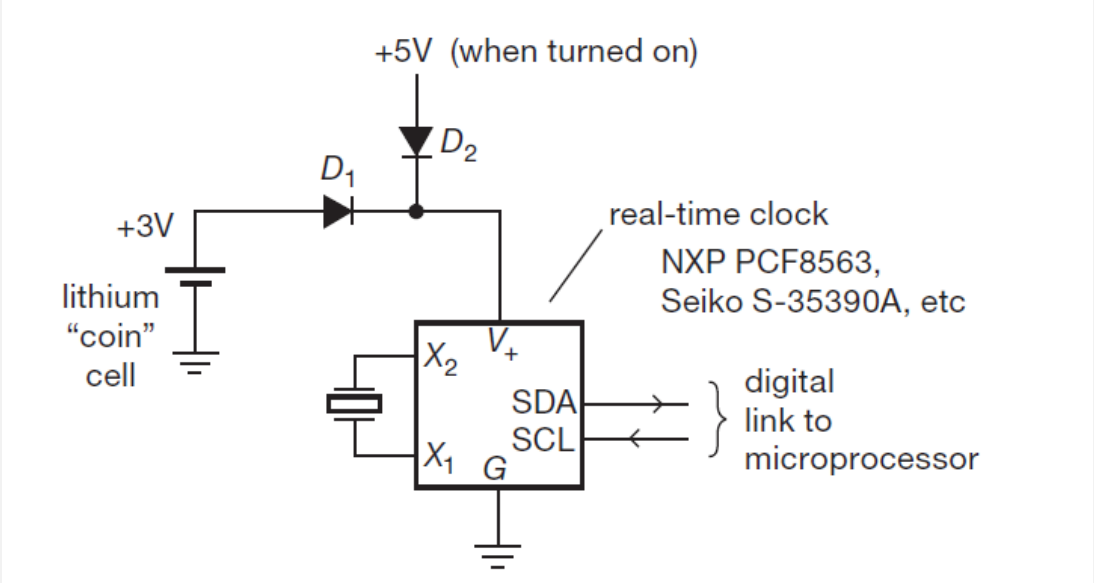
Battery Backup



©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

Aplicații (1)

Diode OR gate

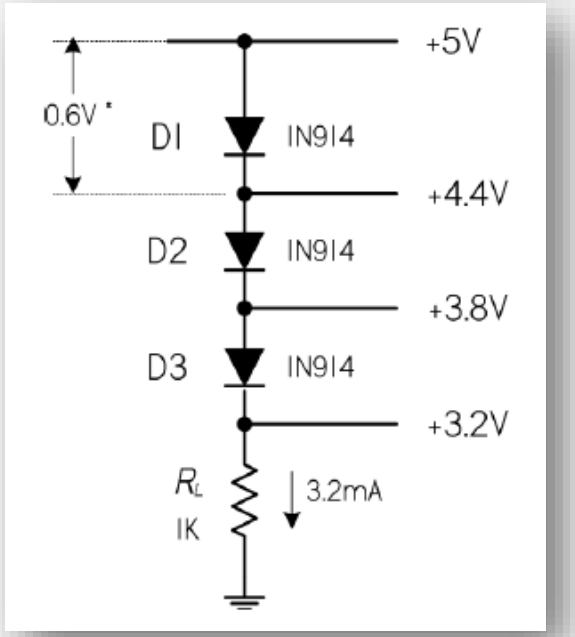


© The art of Electronics, 3rd edition - 2015

Aplicații (2)

Voltage Dropper

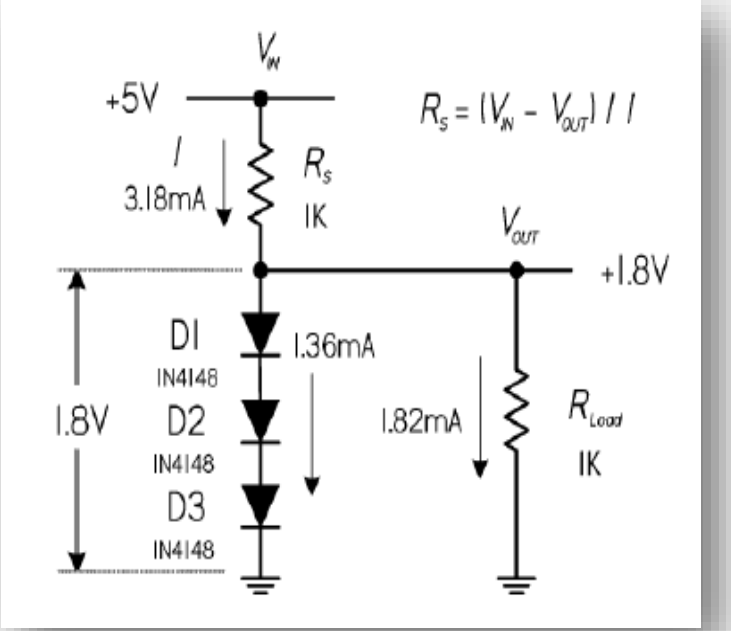
DC Application



©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

Aplicații (2)

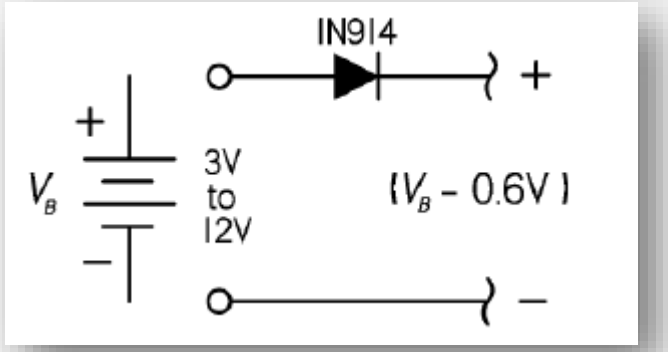
Voltage Regulator



©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

Aplicații (3)

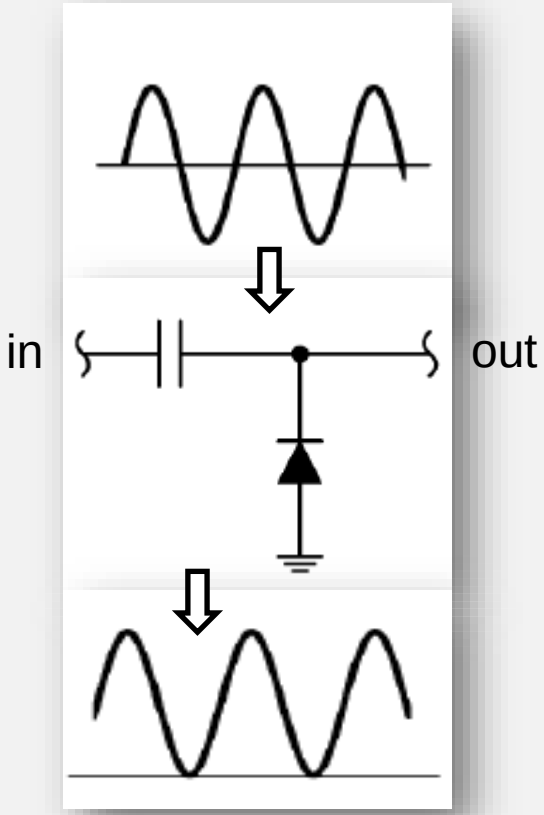
Protecție la alimentare inversă



©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

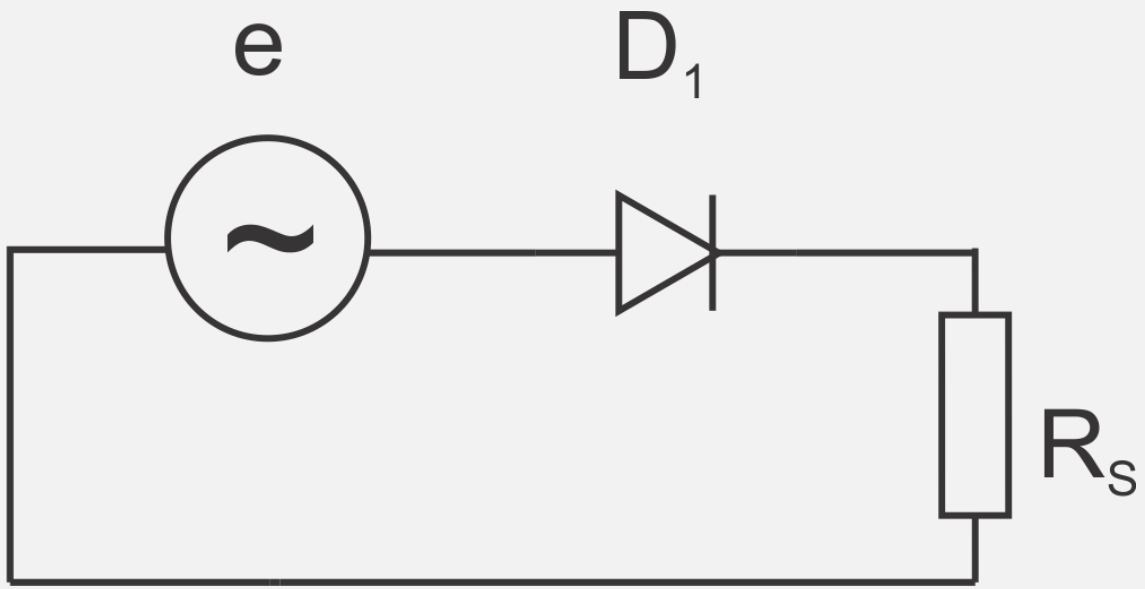
Aplicații (3)

Voltage clamping



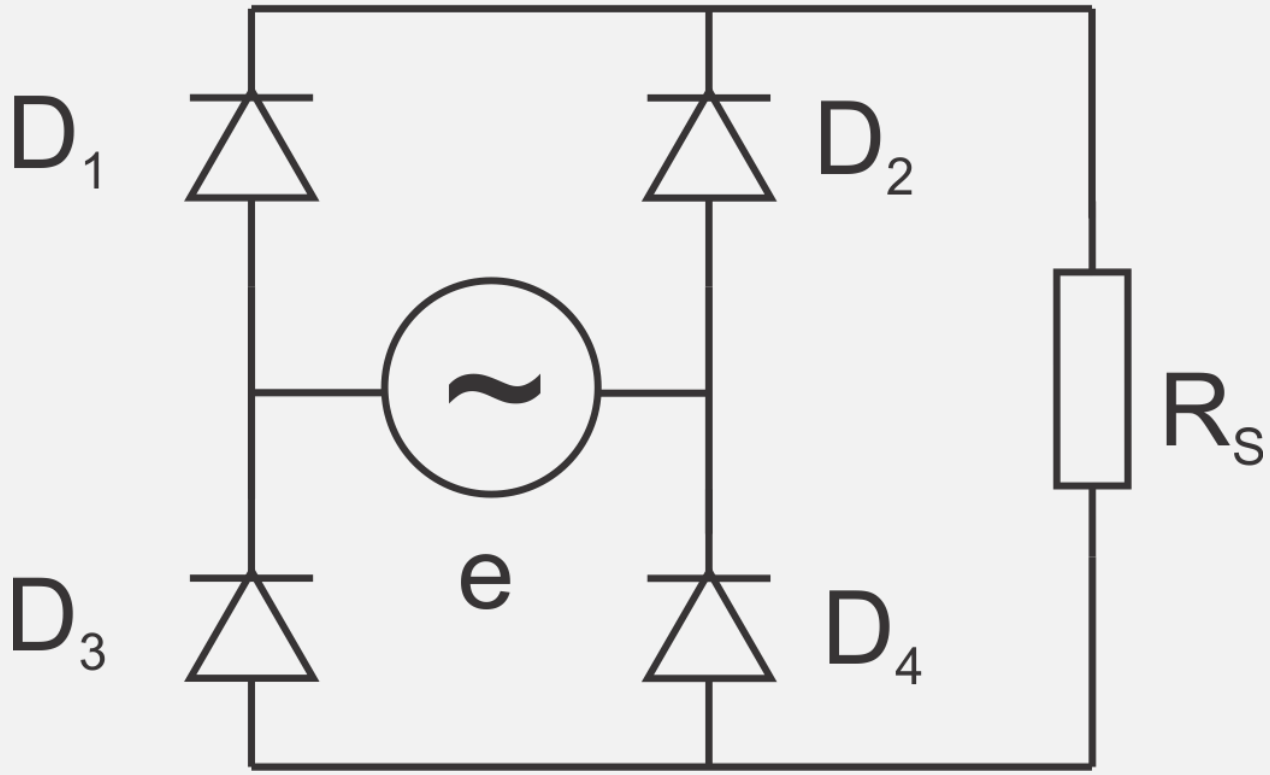
©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

Mono – alternanță

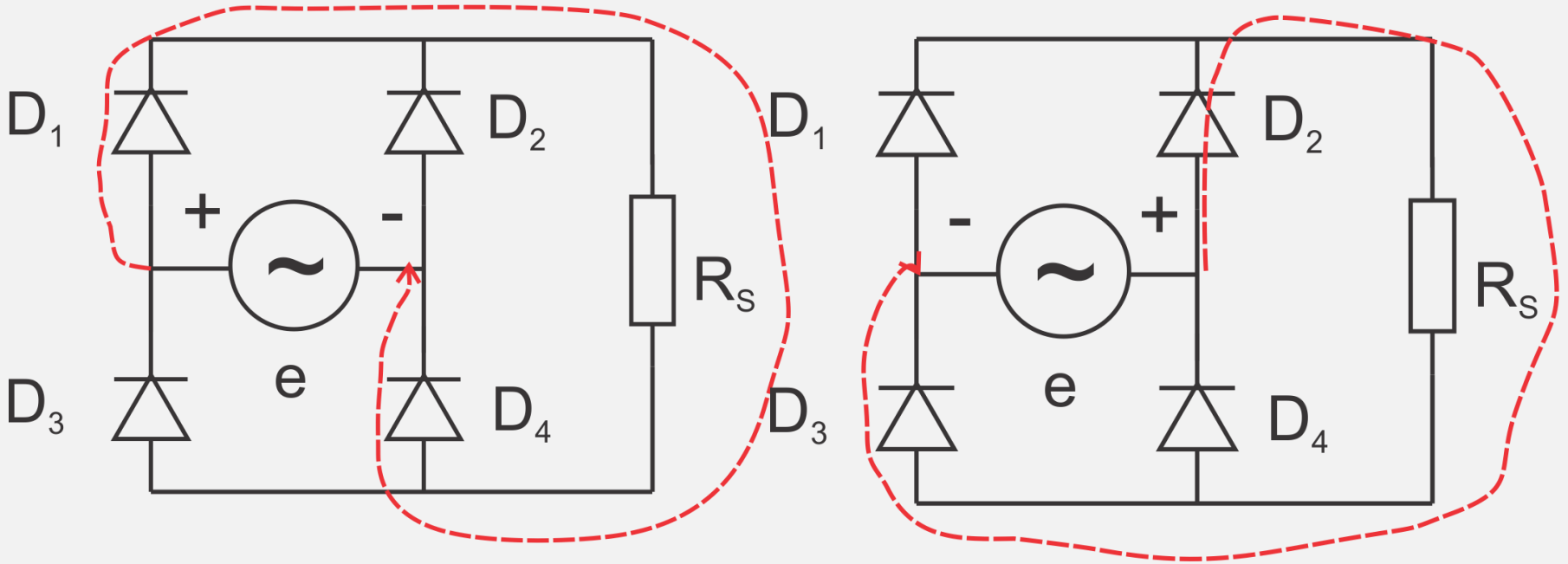


- http://www.falstad.com/circuit/circuitjs.html?cct=$+1+5.0E-6+13.097415321081861+55+5.0+50%0Av+176+336+176+112+0+1+40.0+5.0+0.0+0.0+0.5%0Ar+480+112+480+336+0+640.0%0Ad+176+112+480+112+1+0.805904783%0Aw+176+336+480+336+0%0Ao+0+64+0+35+5.0+0.0125+0+-1%0Ao+1+64+0+35+5.0+0.0125+1+-1%0A

Bi – alternanță (1)

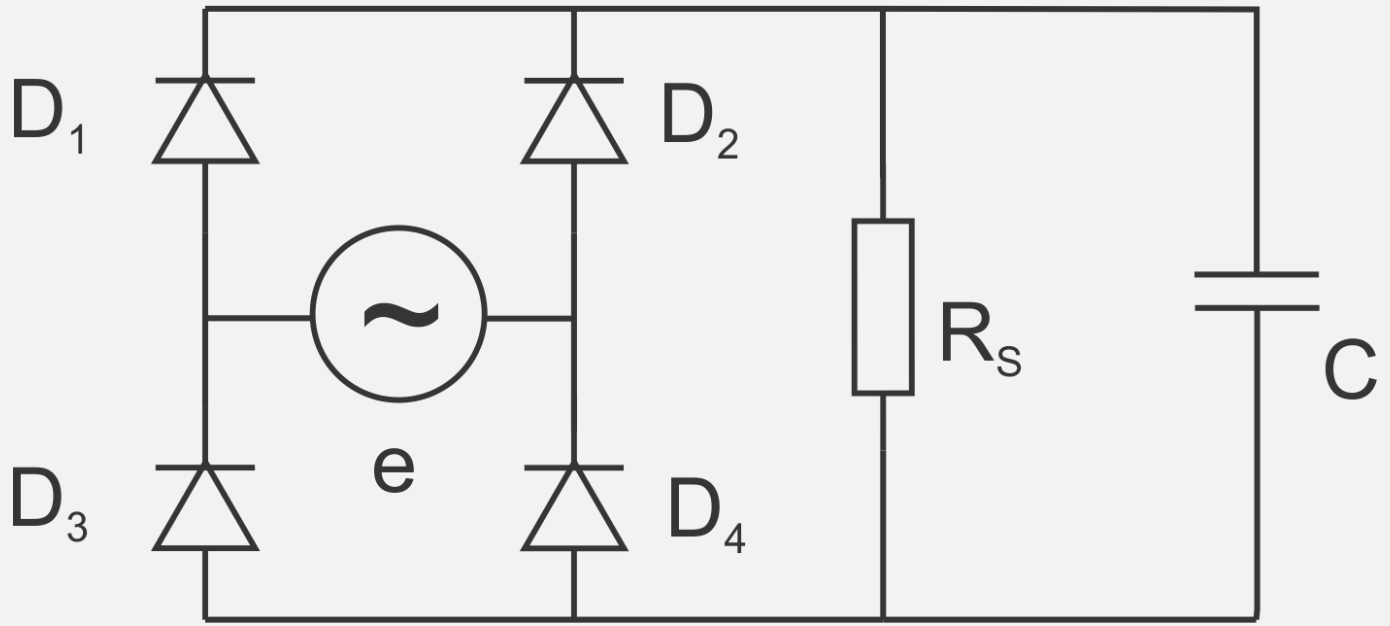


Bi – alternanță (2)



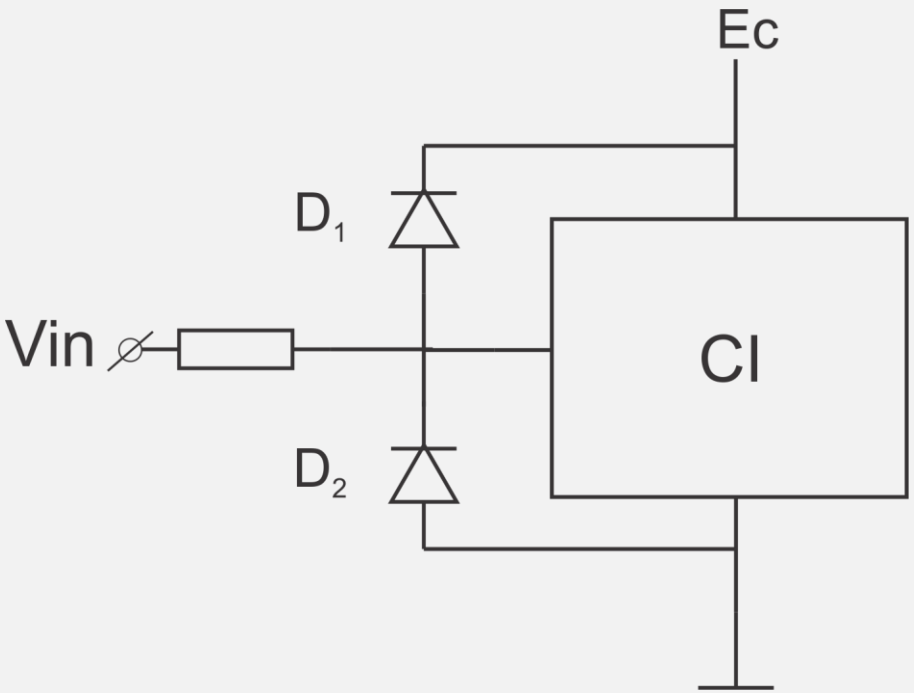
http://www.falstad.com/circuit/circuitjs.html?cct=$+1+5.0E-6+10.20027730826997+53+5.0+50%0Av+544+592+544+304+0+1+40.0+5.0+0.0+0.0+0.5%0Aw+544+304+688+304+0%0Aw+688+304+688+368+0%0Ad+688+368+752+432+1+0.805904783%0Ad+688+496+752+432+1+0.805904783%0Ad+624+432+688+368+1+0.805904783%0Ad+624+432+688+496+1+0.805904783%0Aw+688+496+688+592+0%0Aw+688+592+544+592+0%0Aw+624+432+624+528+0%0Aw+752+432+800+432+0%0Aw+624+528+800+528+0%0Ar+800+432+800+528+0+100.0%0Ax+847+488+884+493+0+20+load%0Ao+0+64+0+35+10.0+0.05+0+-1%0Ao+12+64+0+35+5.0+0.05+1+-1%0A

Bi – alternanță (3)



http://www.falstad.com/circuit/circuitjs.html?cct=$+1+5.0E-6+10.20027730826997+50+5.0+48%0Av+144+368+144+80+0+1+40.0+5.0+0.0+0.0+0.5%0Aw+144+80+272+80+0%0Aw+272+80+272+144+0%0Ad+272+144+336+208+1+0.805904783%0Ad+272+272+336+208+1+0.805904783%0Ad+208+208+272+144+1+0.805904783%0Ad+208+208+272+272+1+0.805904783%0Aw+272+272+272+368+0%0Aw+272+368+144+368+0%0Aw+208+208+208+304+0%0Aw+336+208+384+208+0%0Aw+208+304+384+304+0%0Ac+384+208+384+304+0+1.02E-4+3.595636740813719%0Aw+384+208+464+208+0%0Aw+384+304+464+304+0%0Ar+464+208+464+304+0+430.0%0Ax+499+264+529+268+0+16+load%0Ao+0+32+0+34+5.0+0.05+0+-1%0Ao+15+32+0+35+5.0+0.0125+1+-1%0A

Protecție Circuite Integrate



- http://www.falstad.com/circuit/circuitjs.html?cct=$+1+5.0E-6+30.13683688681966+50+5.0+50%0Ad+416+240+416+176+1+0.805904783%0Ad+416+304+416+240+1+0.805904783%0AR+496+80+496+32+0+0+40.0+5.0+0.0+0.0+0.5%0Ag+496+368+496+416+0%0Aw+416+176+496+176+0%0Aw+496+80+496+176+0%0Aw+416+304+496+304+0%0Aw+496+304+496+368+0%0Ar+416+240+304+240+0+1000.0%0AR+224+240+160+240+0+3+40.0+10.0+0.0+0.0+0.5%0Aw+224+240+304+240+0%0Ar+448+240+496+240+0+100000.0%0Aw+416+240+448+240+0%0Aw+496+240+496+304+0%0Ao+10+64+0+35+10.0+0.025+0+-1%0Ao+12+64+0+291+10.0+3.90625E-4+1+-1%0A

Protecții generale

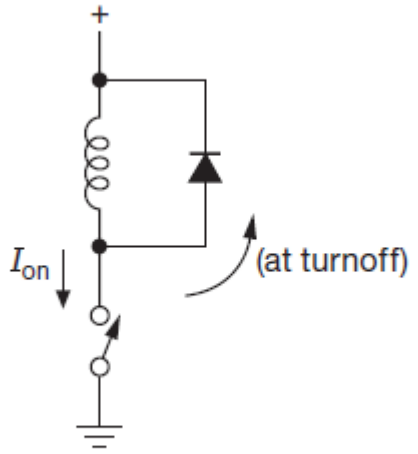
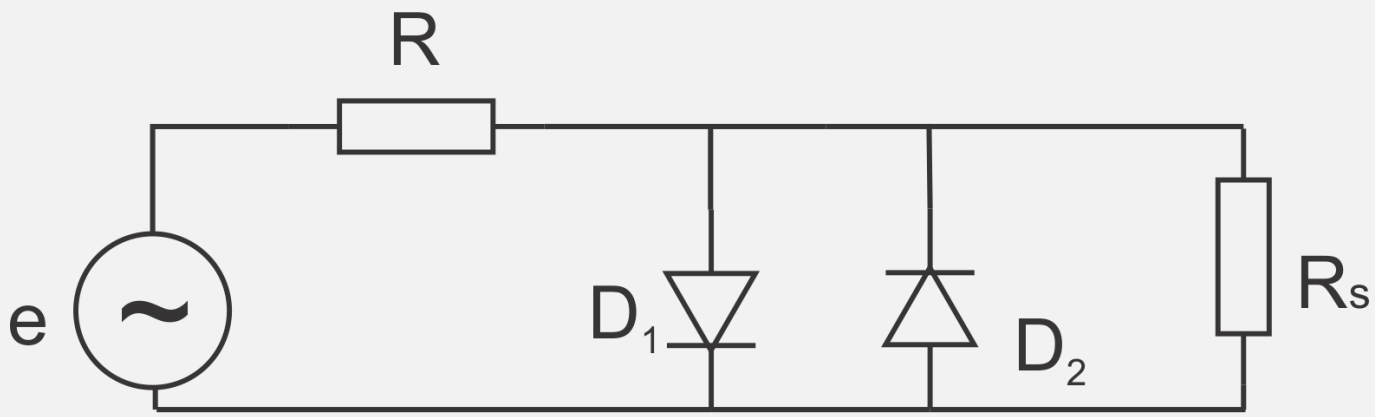


Figure 1.84. Blocking inductive kick.

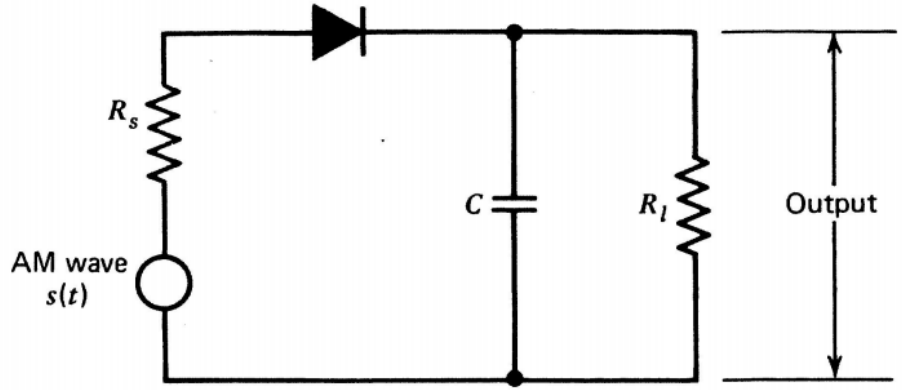
https://tinyurl.com/yymeouyc

© The art of Electronics, 3rd edition - 2015

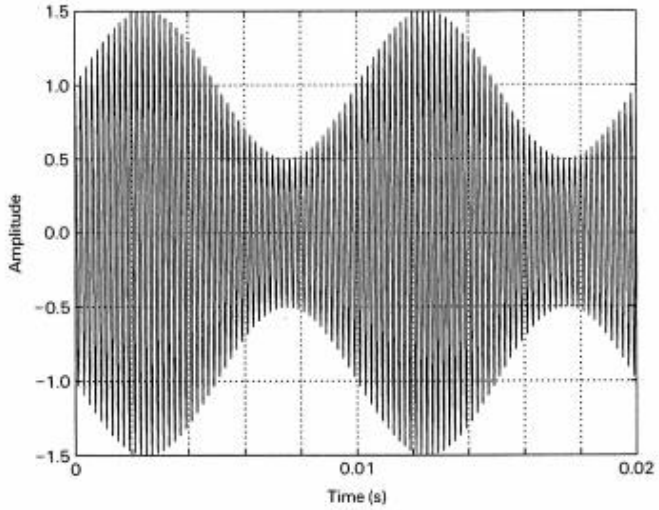
Limitator



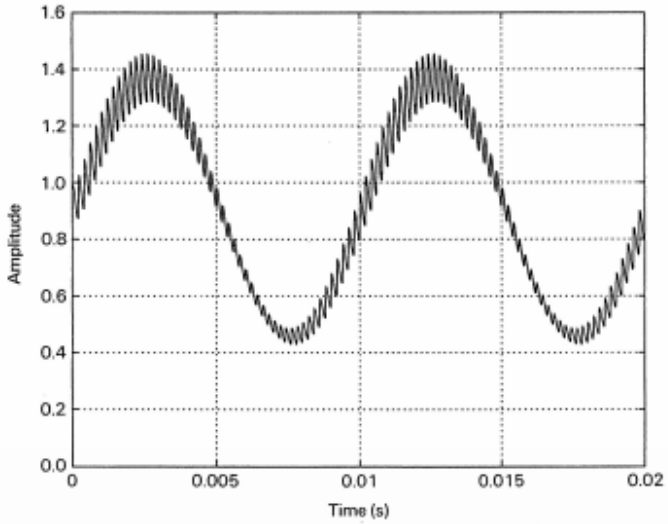
- http://www.falstad.com/circuit/circuitjs.html?cct=$+1+5.0E-6+22.512744558455275+50+5.0+50%0Av+240+304+240+160+0+1+40.0+5.0+0.0+0.0+0.5%0Ar+240+160+336+160+0+110.0%0Aw+336+160+336+208+0%0Aw+336+208+368+208+0%0Aw+336+208+304+208+0%0Aw+240+304+304+304+0%0Aw+304+304+368+304+0%0Ad+304+208+304+304+1+0.805904783%0Ad+368+304+368+208+1+0.805904783%0AO+336+128+432+128+0%0Ar+432+192+432+256+0+1000.0%0Aw+432+256+432+304+0%0Aw+432+304+368+304+0%0Aw+336+128+336+160+0%0Aw+336+160+432+160+0%0Aw+432+160+432+192+0%0Ao+0+32+0+35+5.0+0.05+0+-1%0Ao+9+64+0+290+1.25+2.44140625E-5+1+-1%0A



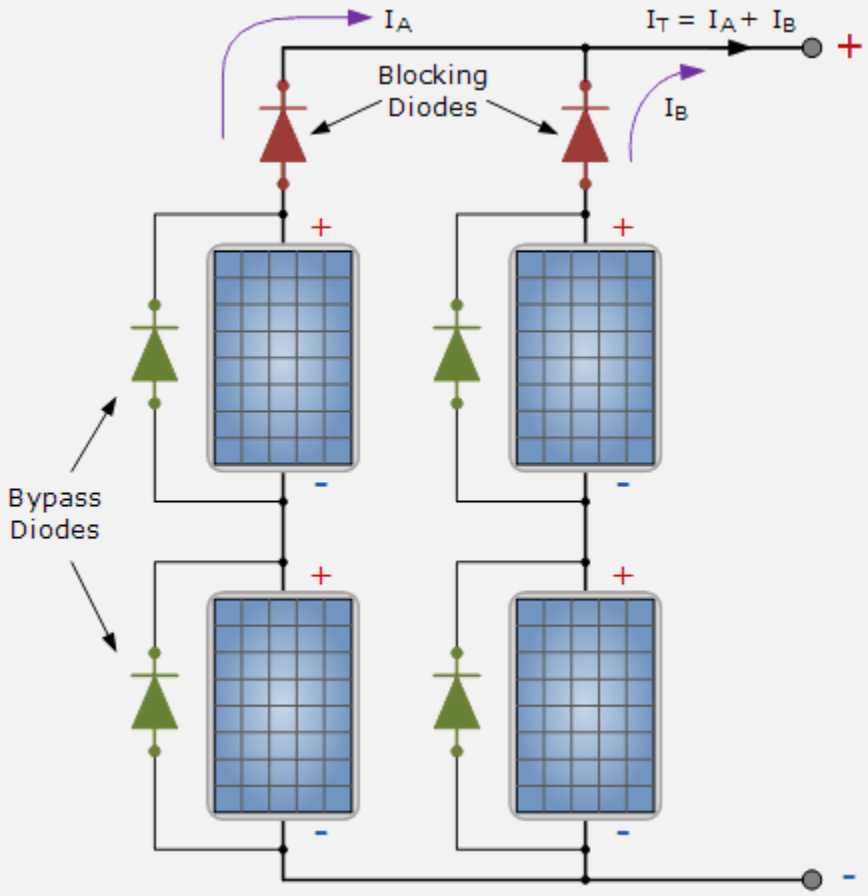
AM wave



Demo-dulated output

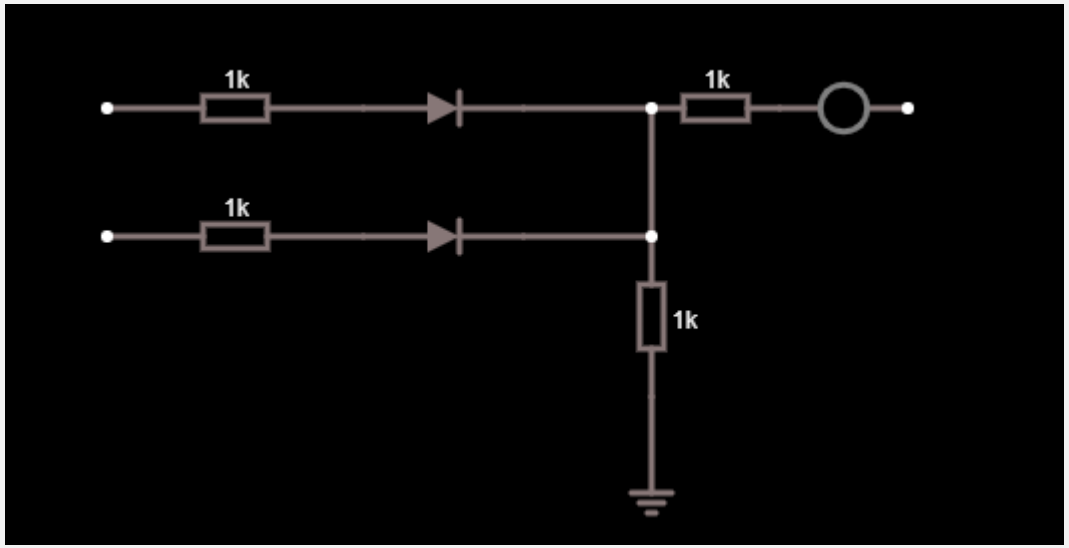


© http://educypedia.karadimov.info/library/tut331_04am_dem.pdf



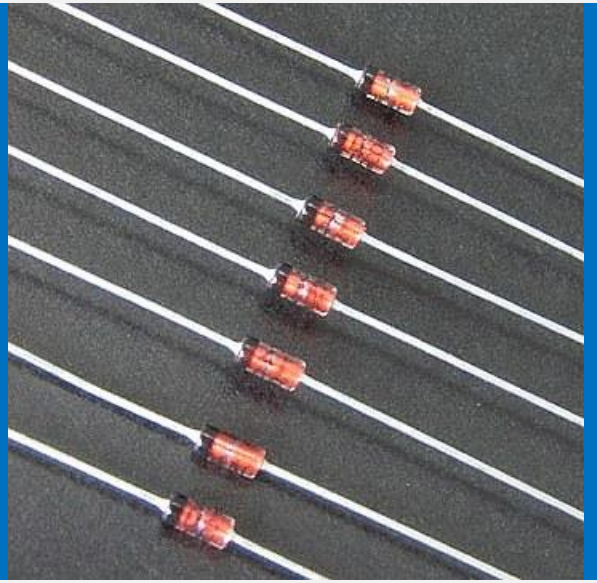
© https://www.electronics-tutorials.ws/diode/bypass-diodes.html

++ in sem 2 – logic gates

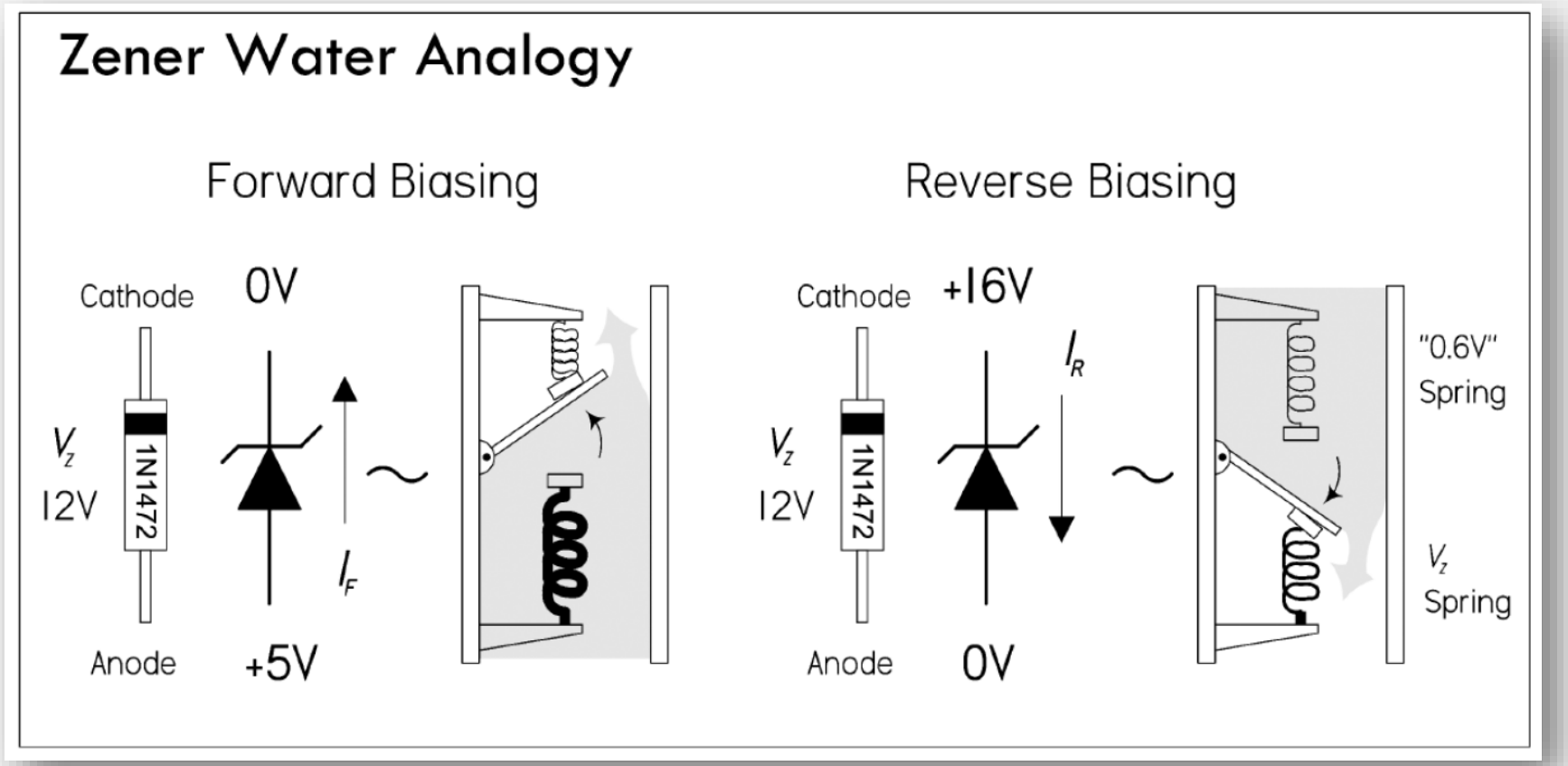


© https://www.electronics-tutorials.ws/diode/bypass-diodes.html

Zener



Analogie Dioda Zener/Apă

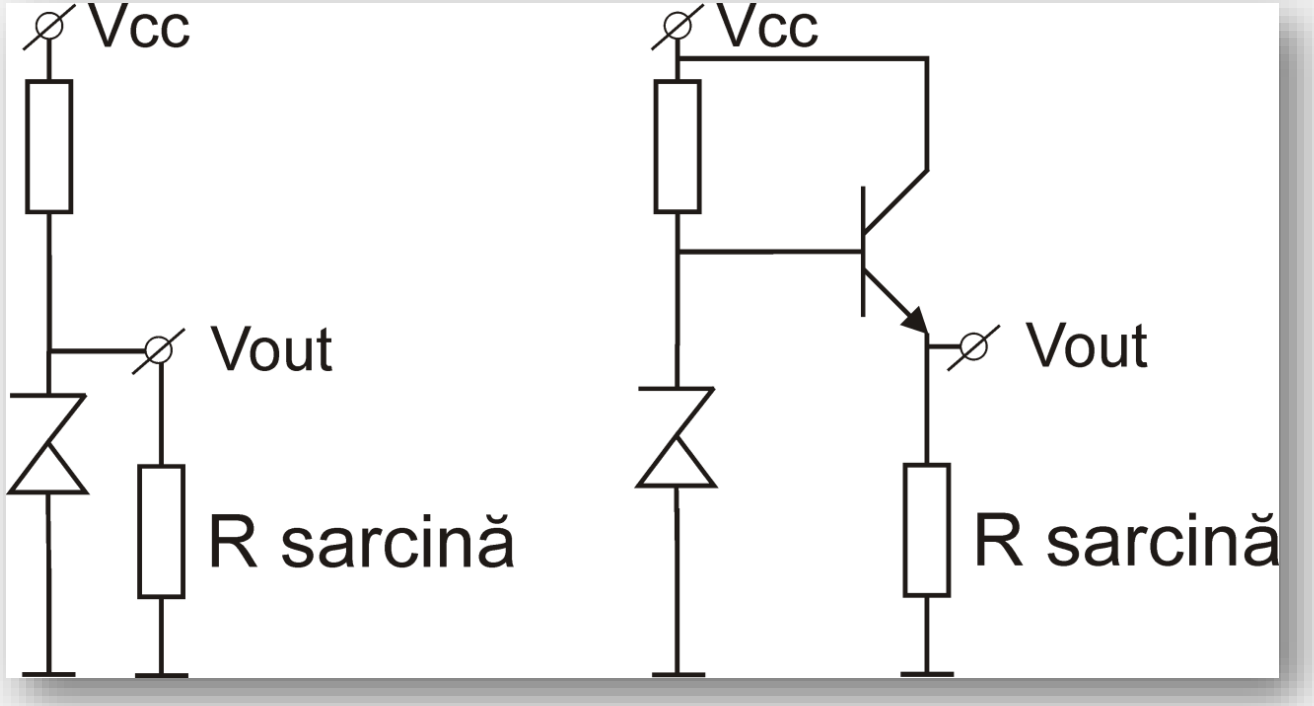


- Dioda Zener - „two-way gate”

©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

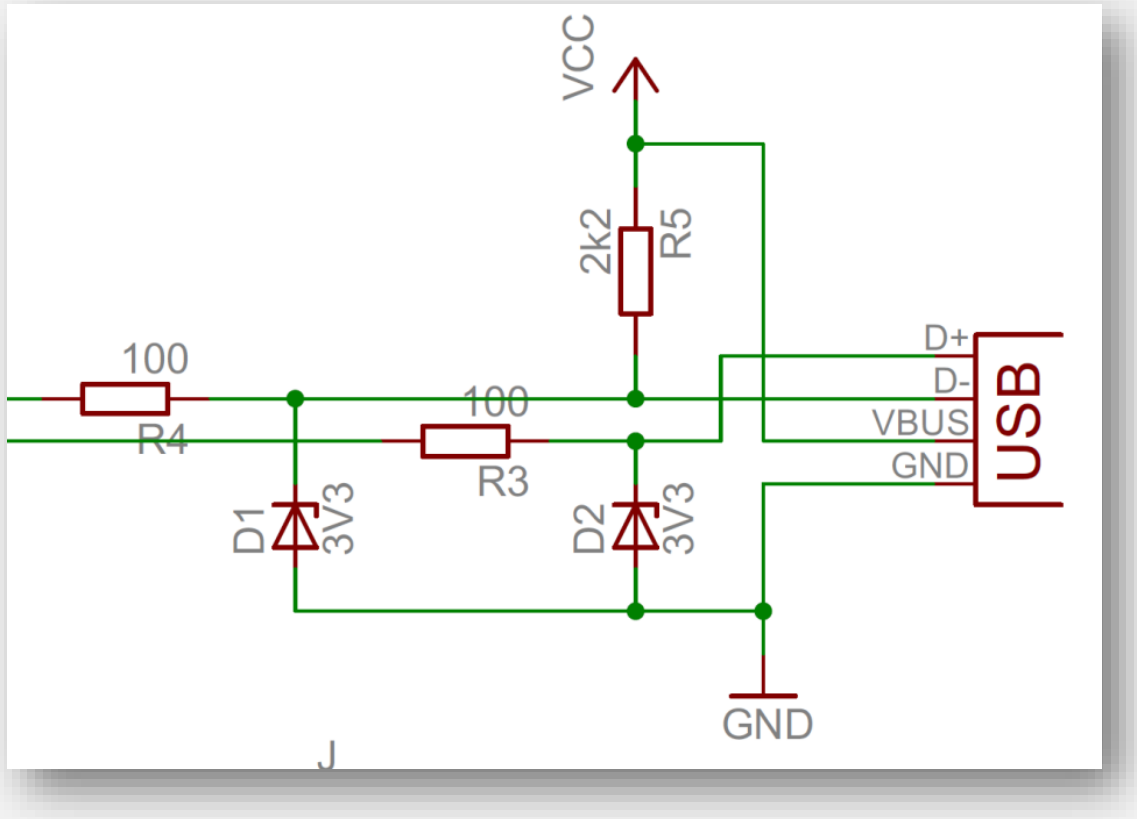
Aplicații Zener (1)

- Atenție la Puterea disipată!
- Simulare la curs: **tinyurl.com/ndjhedct**



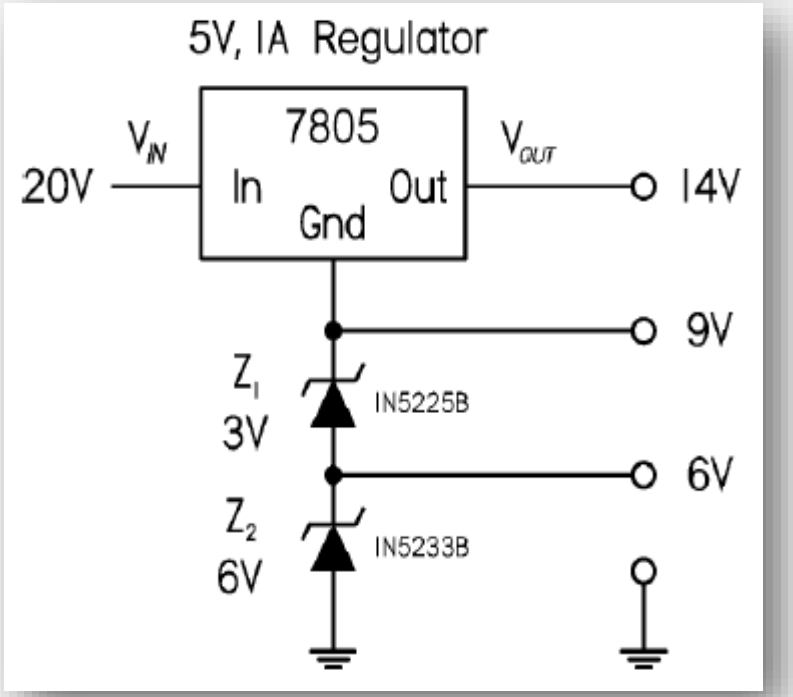
Aplicații Zener (2)

- 3,3V (USB)
- 5,6V (+ NPN) => 5V
- Limitare (2 x Zener)

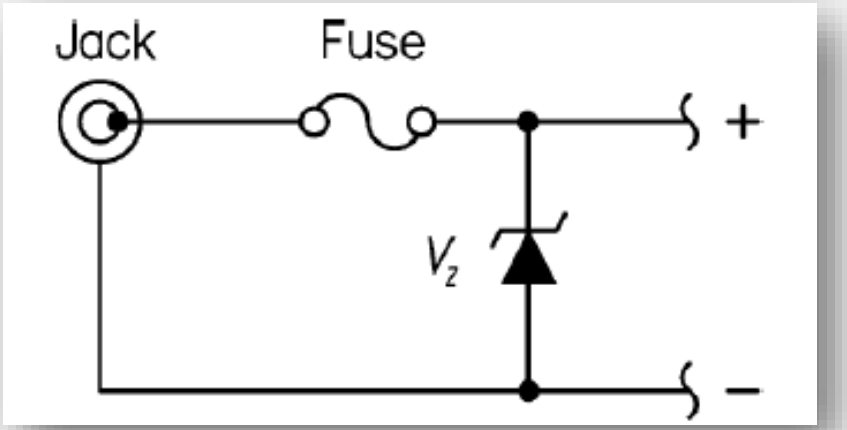


Aplicații Zener (3)

Voltage Regulator Booster



Overvoltage Protection



©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

Aplicații Zener (3)

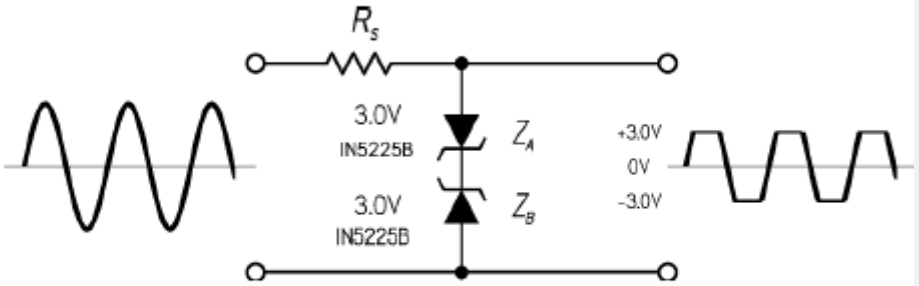


FIGURE 4.32

©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

Aplicații Zener (3)

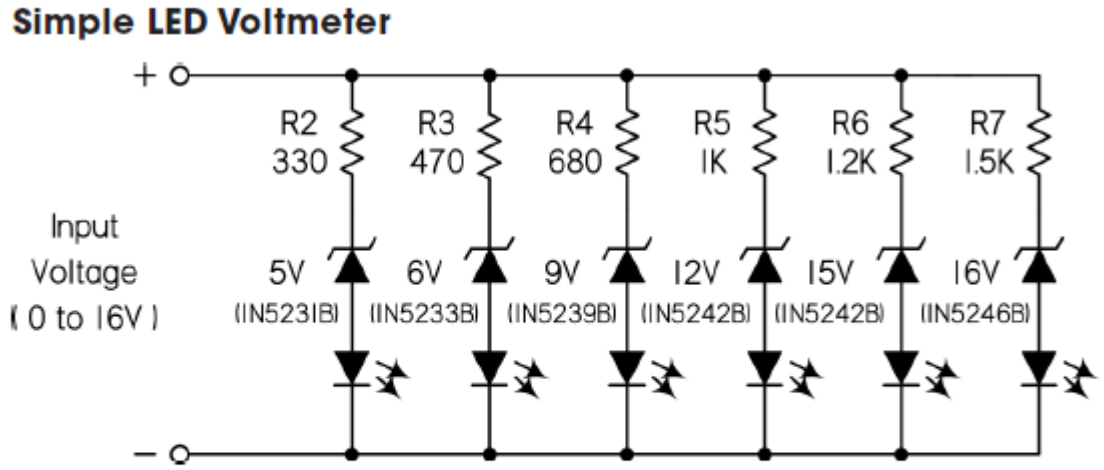


FIGURE 4.37

©Practical Electronics for Inventors (Paul Scherz și Simon Monk, 2016) - imagini

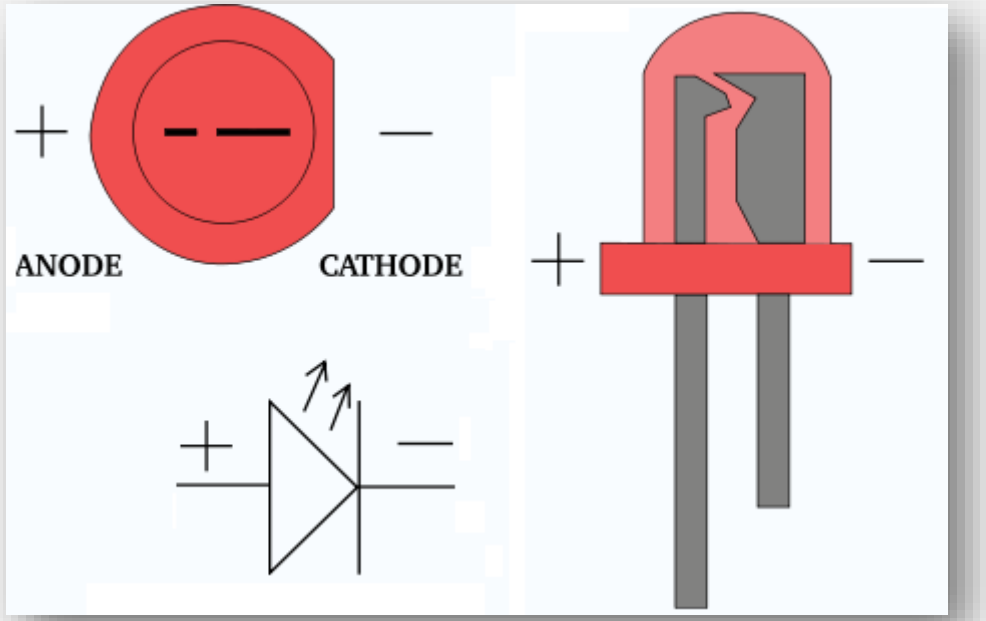
Fotodiode

Diode electroluminiscente



LED-uri (*light-emitting diode)*

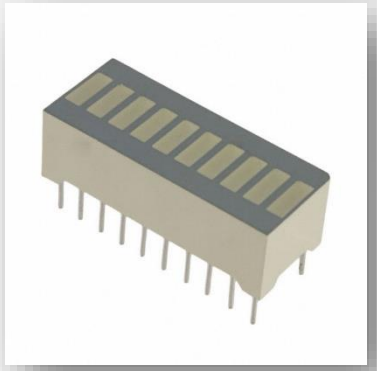
- Diodă semiconductoare ce emite lumină la polarizarea directă a joncțiunii p-n

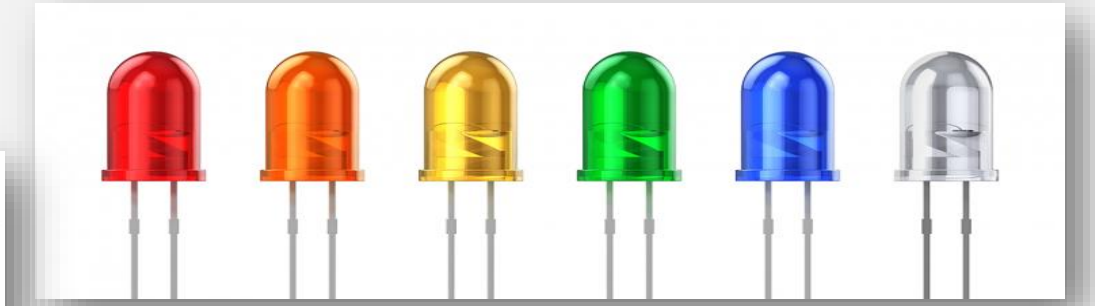


Cădere de tensiune LED-uri

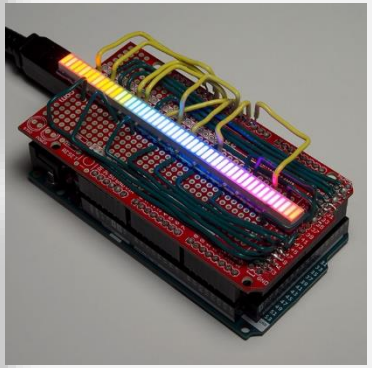
Culoare LED	Cădere de tensiune (V)
Roșu	1.63 – 2.03
Galben	2.10 – 2.18
Portocaliu	2.03 – 2.10
Albastru	2.48 – 3.7
Verde	1.9 – 4
Mov	2.76 – 4
UV	3.1 – 4.4
Alb	3.5

Variante de LED-uri

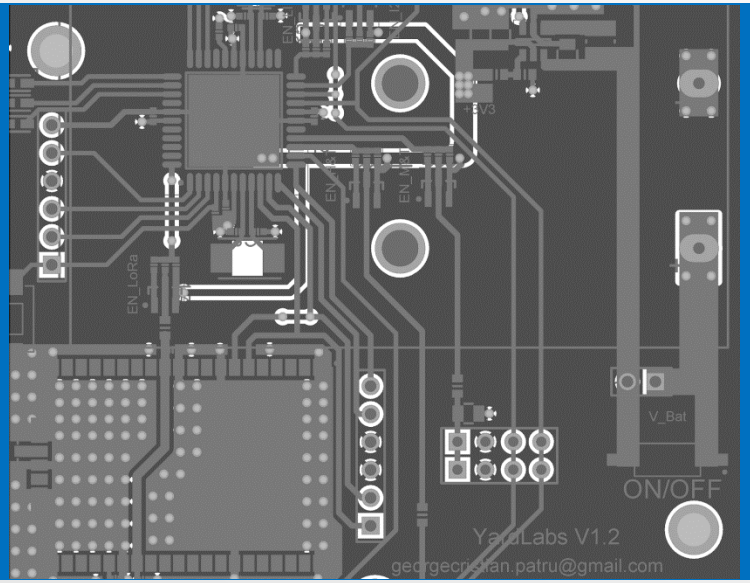








Exemple


EN_LoRa
V_Bat
ON/OFF
YaroLabs V1.2
georgecristian.patru@gmail.com